

ANNEX G: CLOUD DATA GUIDANCE

**Version 5.2
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REVISION SUMMARY

Version	Revision Date	Revision Summary
1.0	12/31/2015	Initial Release
2.0	02/29/2016	Updated storage guidance. Added migration and metadata guidance.
3.0	4/19/2016	Moved storage guide materials to white paper, revised most sections.
4.0	6/30/2016	Minor edits and corrections. Added information on Open Data Store. Moved cost info to another annex.
4.1	08/05/2016	Updated classification/portion marking to support push to SBU
5.0	9/30/2016	Added data movement process descriptions, and chart. Edited migration sections for consistency. Added data conditioning explanation. Revamped S3 Example Case. Added Database Example Case, and S3 Bucket Naming Standards.
5.1	1/6/2017	Added data access information. Changed 'Data Owner' to 'Data Custodian'. Revised migration information. Extensive revision to Appendices.
5.2	5/31/2018	Revised A&A and data to the cloud procedures. Replaced references to Content Accounts with references to Open Data Store.

Annex G: Cloud Data Guidance

1.0 INTRODUCTION

This Annex provides cloud data guidance, establishes a common terminology for data service offerings, provides data migration guidance, and describes how NGA will manage its metadata in the OpenDataStore(ODS) cloud. Key data requirements and recommended best practices are identified for programs adopting cloud services. Cloud data guidance is guided by the NGA Unified Data Strategy goals outlined in Table 1 below.

Table 1: NGA Unified Data Strategy Goals

<i>Goals</i>	<i>Definitions</i>
<i>Find the Data</i>	Users across NGA, the IC and DoD are able to intuitively discover, access, and retrieve the geospatial data and content
<i>Free the Data</i>	Data are required to be separated from individual applications, systems and information silos
<i>Protect the Data</i>	Data accessibility is based on user role and access control data attributes, reflecting user rights, roles and data usage; available when and where needed
<i>Automate the Data</i>	Data are conditioned with metadata throughout its lifecycle to enable automated access, discovery, pedigree, provenance, retrieval, and analysis
<i>Unify the Data</i>	Integrated Intelligence is unified across all boundaries: architectures, security domains, geographic locations, and mission, corporate and functional operations

As NGA migrates activities to cloud environments, data guidance will help define the data architecture, tools, and processes required to achieve success. Overarching concepts are that NGA data will be:

- Managed in accordance with NGA data management, retention and records management policy,
- Encrypted in transit and at rest,
- Available to the National System for Geospatial Intelligence (NSG) and Intelligence Community (IC),
- Cataloged and indexed for discovery (e.g., using Enterprise Catalog and search engine services)
- Tagged with handling and access information, and
- Exposed via services.

There are distinct intranets serving the IC and DoD communities, and the classified domains are air-gapped. The cloud in this guidance includes:

- Top Secret Cloud (TC)
- Secret Cloud (SC)
- Unclassified Cloud (UC).

In addition, there are interrelated IC cloud resources being the IC GovCloud and IC ITE. The data storage and data services portion of the latter is in the TC's OpenDataStore.

The key concepts introduced will enable the reader to better understand the guidance for: ODS, Data Access, Analytic Data Lake, Conditioning, Data Migration Categories, ContentID, and Roles.

Information is provided on key concepts and Chief Data Officer (CDO) direction on how to handle data and metadata in the cloud environments, particularly in the OpenDataStore.

1.1 OpenDataStore

The ODS represents an architectural evolution in data management. In the past, data belonged to and was stored and managed by applications often in semi-private data structures. With ODS, data will be managed by enterprise data services and maintained in common areas accessible to all authorized users, as illustrated in Figure 1 and further described in the Design White Paper on Data Architecture.¹

The ODS contains both persistent storage of data accessible to end users for normal work and analysis, and data services. It is a highly inclusive concept covering all data stored for normal end-user use. It encompasses all means and methods to store data, from flat files, to relational databases, NoSQL databases and complex structures like data warehouses and Analytical Data Lakes. It also includes all formats, from structured relational, to object, to triple-store, to unstructured files.

ODS is an amalgamation of two concepts: Open Data and the Data Store. A Data Store is an inclusive repository, and the Open Data movement advocates that data should be freely available to everyone to use.

Data can be stored in UCs Amazon Web Services (AWS) or TCs Commercial Cloud Services (C2S) S3 buckets or in file system implementations on C2S Elastic Block Services (EBS) raw data storage area. Relational and NoSQL databases may be supported by C2S Relational Database Services (RDS) or with C2S Dynamo. Other services can be set up in an C2S Elastic Compute Cloud (EC2) instance with EBS, as shown in Figure 2.

Data transformation and conditioning services are among the many types of data services that NGA can stand up. While the Open Data movement is about making data free and open to the public, that cannot be done with classified and controlled

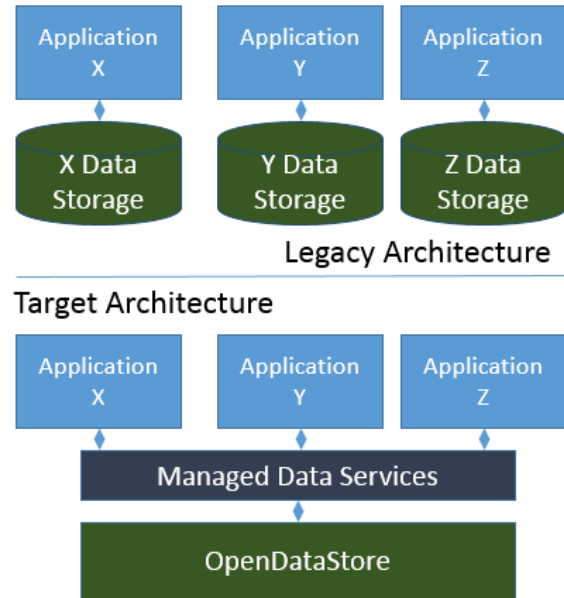


Figure 1: Legacy and Target Data Architecture

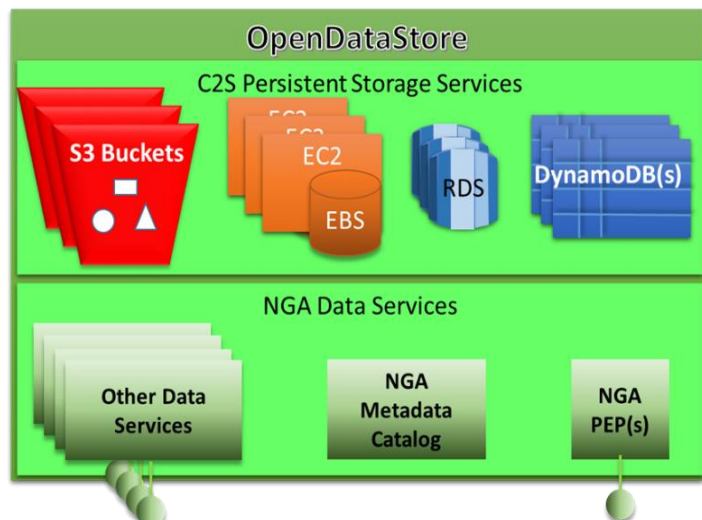


Figure 2: OpenDataStore Services

¹ [Cloud Data Storage: Data Architecture Pattern Choices](#), 2018.

information. The key elements for implementing open data services are:

- Automate the access via Policy Enforcement Points (PEPs) so if the user credentials meet the metadata attributes, the user can access the data.
- Eliminate technical barriers (i.e., organization by application, and use of private data formats) to mission data, beyond that required by the classified networks.
- Design data access services that meet user needs, including services that allow dataset wide discovery and access when appropriate.

The ultimate objective is to provide all data as a *Managed Service*. Initially, this will take the form of providing a shared environment to store data via the OpenDataStore. The ODS will work much like the NGA enterprise file storage and enterprise database clusters of the NGA Data Centers:

- The data infrastructure will be set up for the Application Service Provider (ASP) programs with:
 - Data infrastructure access rights and networking
 - Actual instances for ASP data storage (database or file storage)
- The ASP will be able to configure that data storage instance, and:
 - Define role based access controls internal to their storage solution
 - Configure their storage solution to support their activity needs

One of the goals of the CDO is to ensure that the mission and activity capabilities are not adversely affected by the cloud data migration. As such, initially the CDO will work to enable activities to access their data the same as in the past, including direct connections to data storage.

Over time, the CDO, working with the ASPs, will work to transition all data access to web-based services instead of application-direct access to the data stores. Eventually, the ASPs will be consumers of the data services instead of working directly with the data stores.

1.1.1 ODS Content Accounts

The CDO is creating an ODS account for mission data and a corporate ODS account of corporate data for each security fabric on C2S. All mission data and corporate data will be stored in the ODS.

While other data categories are not required to go in predefined ODS accounts, there is nothing preventing them from being stored there, should the Data Custodian (DC) so choose. There are many instances where activity data shares space with mission data in a common database, and the DC may prefer to store the entire activity data in one account.

1.1.2 ODS Service Providers

There are three (3) service provider roles with respect to the ODS, as identified in Table 2.

Table 2: Service Provider Roles and Responsibilities

<i>Provider Type</i>	<i>Rights & Responsibilities</i>
Infrastructure Service Provider (ISP)	• Creation of AWS accounts. • Configuration/Management of the AWS Identity Access Management (IAM) services in the AWS console in any given account. • Configuration/Management of the network services in the AWS console in any given account.
Data Service Provider (DSP)	• Management of the ODS data storage and data services. • Configuration/Management of all the AWS services in the NGA content. AWS accounts except those managed by the ISP.
Application Service Provider (ASP)	• Create/Read/Update/Delete (CRUD) operations in the various data stores needed by their activity. • CRUD operations against the NGA Metadata Catalog. • Configuration of database accounts for databases related to their activity. • Configuration/creation of database structures and database code. • The ASP will not have the right to access the AWS console, or to spin-up or spin-down instances. They will just be able to access and perform whatever actions granted to them against the specific instances they need to access.

1.1.3 ODS Assessment & Authorization (A&A)

The ODS extends the concept of the shared security model under which C2S operates, such that:

- C2S (i.e., Amazon Web Services) has the responsibility for securing the hardware and virtual machines.
- ISP has the responsibility for NGA ODS accounts and ensuring IAM and network security.
- ASP has the responsibility for the A&A of their activity, the storage service they are operating, and configuring within accounts, databases and other ODS aspects.

Security has stated that activity boundaries do not align with ODS C2S account boundaries, and that the activity A&A includes all parts of the activity architecture, not just the parts within their own C2S account. Therefore, since the ASP is configuring their actual data instance in the ODS, they are responsible for showing that the data configuration meets ICD 503 requirements.

Moving a database from an NGA account or Virtual Private Cloud (VPC) on C2S into the ODS VPC on C2S will not impact or invalidate the current A&A of an NGA activity, so long as the basic environment controls are the same. As these environment controls are set up and controlled in the same configuration for all NGA accounts, this should not be an issue.

Figure 3 illustrates two A&A examples. In the first example, the Target Management System (TMS) activity has an ODS account where web and application server instances reside. The TMS database is on an EBS backed EC2 instance in the ODS mission account, so the TMS architecture spans two accounts, and the TMS activity is responsible for the A&A for its entire architecture.

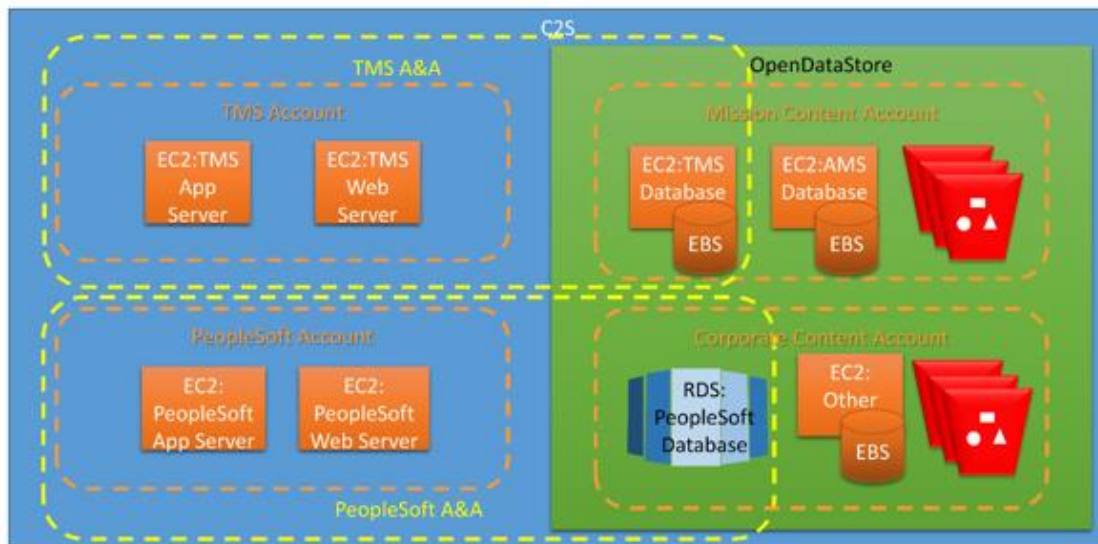


Figure 3: A&A Examples (TMS and PeopleSoft)

The Data Service Provider (DSP) would spin up the TMS instance, using the standard Oracle with a spatial Amazon Machine Image (AMI) and provide the secure access credentials. The activity would then set up individual developer staff logins within that instance and any other configurations required to be ICD 503 compliant in order to achieve Authority to Operate (ATO), and that Oracle database instance. Since they are using the standard AMI, ATO is bounded by the ASP activity.

In the second example, the PeopleSoft application and web servers are in a PeopleSoft account, but the PeopleSoft database is on the RDS managed service in the ODS. The DSP stood up the RDS instance for the ASP, but the ASP is responsible for setting up their schema, setting up developer accounts in Oracle along with other infrastructure tasks. The ASP is still responsible for the A&A of the entire PeopleSoft architecture, even though it spans the PeopleSoft account and the ODS.

In both cases, there will have to be test cases that prove that the ASP NIST 800-53 controls are satisfied. In summary, the ISP, the DSP and the ASP will each have NIST 800-53 controls that will need to be satisfied through ATO test cases. The controls can be unique to a service provider, or shared as negotiated with CIO-T. However, A&A boundaries are not determined by hardware nor account boundaries. For additional information on the cloud A&A process, see the Cloud Security Annex.

1.1.4 ODS Exceptions

The vision for the ODS is to support the data needs of all NGA missions and activities. However, it is understood that there may be exceptions based on security, policy or technical needs. For activities that need more rights with respect to their data, or data that needs greater protections than are possible in a common ODS account, the CDO will work with the activities and the DC on a case-by-case basis to determine what is needed to resolve their issues, and grant exceptions if necessary.

1.2 Data Access

Data access has the following control constraints and features, per DNI Directives:

- There is an automated way to discover and access data. All data is metadata tagged, person identity and roles are associated with accessibility attributes, and automated rules and processes use these tags to automatically enforce accessibility and dissemination policy.
- “Responsibility to provide” is achieved. The default stance on data is that it is shareable, unless there is a specific dissemination control, exemption or constraint preventing it from being shared. However, “responsibility to provide” does not negate our responsibility to protect sources and methods.
- People with the “appropriate security clearance and an assigned mission need” are able to access data. Entitlements are how “need to know” is implemented. No other “need to know” check is performed – it is implicit in the request.

1.3 Analytical Data Lake

An Analytic Data Lake is a popular approach for analysis of large amounts of data, in that: *“A Data Lake is an attempt to bring together physically available data stores in such a way as they can be easily accessible to users.”*² Data Lake characteristics of include:

- Data is sub-transactional (append only) or non-transactional (read only)
- Data is used by multiple user communities
- Data is of a scale/volume that won’t economically fit into a relational database
- Data is in a native format (i.e., raw)
- A Data Lake is an analytical tool. It allows the user to put massive amounts of disparate data in one spot and run analysis across all of it. It is not, however, a data repository for all data at NGA. For example, the production databases for Foundation GEOINT data are updated thousands of times per day, so these would not be able to exist in an “append only” or “read only” structure.

A Data Lake might not be permanent, in that, it would be created to just answer a specific question. The data sets needed would be loaded, the question asked and answered, and then the Data Lake taken down. So, rather than one, massive, permanent Data Lake at NGA, there could be many Data Lakes, created and taken down in response to analytic need.

- The implementation of a Data Lake would be via a Hadoop, Spark or an Accumulo cluster set up in C2S. Also, note that the data risks becoming non-discoverable if not properly conditioned and metadata tagged.

1.4 Conditioning

Conditioning is the formatting and metadata processing (i.e., cataloging and tagging) of ingested data. Ingest conditioning is defined as that metadata and formatting work minimally required to allow successful load into persistent storage, including the Enterprise Data Header (EDH) handling information. The scope of conditioning is ‘just enough’ to achieve ETL processing without impeding the load or causing unnecessary rejection of the ingest data. Conditioning may continue to be performed at a later time to include:

² *The Confusion and Hype Relating to Data Lakes*, Gartner, 2014.

- External metadata enhancements (i.e., populating metadata catalogs, search engine indices, and semantic web links).
- Contextual metadata improvements (i.e., database metadata, object wrappers and headers, and in-line tags, such as XML micro formats).

1.5 Data Migration Categories

Data migration categories include:

- **Mission Data:** Data used in performance of the NGA mission, which typically would be GOLD copy or record data. This would include source data (e.g., host nation publications, imagery, commercially purchased datasets), the actual production data stores (e.g., Structured Observation Management (SOM), Aeronautical Migration System (AMS) and NGA products (e.g., Digital Nautical Chart (DNC), Intel Reports, Geodesy Models, Navigation Planning (NAVPLAN) Charts, and Fleet Guides).
- **Activity Data:** Data (e.g., code, configuration files, logs) other than mission content, needed to make an activity/service run.
- **Analytical Data:** Ephemeral working data/products that analysts are analyzing on their desktops or in an analytical tool/service. It is not considered mission data until it is finalized/published to the repository and/or the customer. The published product created from the results of analytical data is mission data.
- **Corporate Data:** Data used at the enterprise level that enables the business of NGA, but is not directly part of the mission. This would include: PeopleSoft data, Identity and Access Management (IdAM) data, Personal Identifying Information (PII), financial systems, enterprise services, workflow and process tools.
- **Organizational Data:** Data needed for the business process of individual organizations. This would include: International Standards Organization (ISO) processes, organization charts, briefings, and data that organizations put in shared organization drives.

1.6 Content-ID

Content-ID is a way to categorize a related set of activity data, for example: “Digital Nautical Charts” or “Absolute Gravity Survey.” A Content-ID is a unique identifier created when a logical set of activity data is registered in the ATLAS’s Data Inventory. Each data set identified by a Content-ID has a DC and Data Steward (DS), retention/lifecycle policies, and access control policies. The Content-ID is used to track data throughout the migration process. Virtual folders within S3 buckets are created based on Content-ID, and access control and lifecycle policies are then applied to those Content-ID virtual folders based on the rules for that logical set of data.

1.7 Roles

Data is acquired, managed and provisioned through various user roles, as identified in Table 3.

Table 3: Data User Roles and Responsibilities

User Title	User Role
NGA Chief Data Officer (CDO)	Responsible for Data Governance at NGA in general, and for all ODS data and accounts in particular. The CDO ensures that the NGA ODS accounts are properly maintained and configured. As the owner of the C2S CDO Mission Data and Corporate Data, the CDO must approve any new Data or Data Services being added to the Account(s). The CDO also covers direct data migration costs, to move data to the cloud.
Data Custodian	Responsible for bringing the data into the agency. Often known as the Data Owner, this might be the person who: • Authorized the expenditure of funds to buy a set of data • Authorized an Analyst to retrieve a set of free data • Approved a deal with a host nation or other entity to obtain data • Is in charge of the group that produces the data/product One way or the other, the person, with the authority, that is responsible for that set of data at NGA. The Data Custodian is responsible for the ongoing maintenance of his/her data. This includes budgeting for storage costs (including in the cloud), paying storage bills, setting retention policies on the data, and making sure data sets are kept up-to-date or retired as appropriate. Data Custodians must approve of every data set belonging to them being migrated so that they are aware of the scope of their data and can execute their responsibilities.
Data Stewards	Charged with making all information collected and all analysis produced by an IC element available for discovery by automated means by authorized IC personnel ³. In doing so, they: • Define and manage access and handling policies for their data set so that the data is treated in accordance with applicable legal and policy constraints. • Issue Data Management instructions throughout the data life cycle. • Determine if the existing cloud services provide access control mechanisms that meet their security needs.
Data Technical Staff	Perform technical actions on behalf of the data steward. For example, they may prepare and route data to its intended destination in the desired format with appropriate Smart Data labels applied. Or, perform metadata tagging research. ⁴
Data Knight/Champion	Often the person delegated to attend the DMWG and coordinates with the Data Custodians and Data Stewards in their organization to answer data related inquiries.
Activity Data Manager	This role is assigned to one or more Activity automated systems, not a specific user. Gives authorized system users rights to manage (Create, Read, Update and Delete) that Content on behalf of the Data Custodian. Systems that are not Activity Data Managers will only be given Read access to data. For example, the AMS Activity is the assigned Activity Data Manager for the AMS Database per the AMS Data Custodian. If AMS were not, the AMS software could not be used to insert, update and delete data in the AMS database.
Data Scientists	Determine the relevancy of data with respect to mission need and investigate and prototype data analytic approaches to yield useful analytic conclusions from one or more IC data sources. They may provide input into the indexes, extractions, or correlations to be performed against their submitted data. Note that this role is intended to also cover roles of other IC users (e.g., data specialists, developers) who perform similar activities aimed at extracting knowledge from data.
Developers	Design, develop, and implement services and data flows to support needs.⁵ They interact with/use other services to develop, integrate, and deploy capabilities. May also be known as software solution engineers and by other terms.

³ [IC Directive 501, Discovery and Dissemination or Retrieval of Information within the IC.](#)

⁴ Defined in ODNI ES 00594 as “An organization that is responsible for executing data-related tasks on behalf of an originating element. These tasks may include collecting, tagging, and processing data.” Note that the Data Custodian organization may be different than the originating organization.

⁵ [IC ITE Data-Centric Management Principles, v02, 2013](#)

For a user role(s) example, consider the NGA Aeronautical Safety of Navigation group (SFA) DAFIF. A DAFIF Program Manager (PM) is in charge of the product line. The mission DC for DAFIF would be the head of SFA, and the DAFIF PM would be the Data Steward.

1.8 Data Inventory

The NGA's Data Inventory is maintained in the [ATLAS](#) and contains consolidated information collected about NGA cloud migration data⁶. The ATLAS Data Inventory contains:

- Data status information: Data Status, Migration Status.
- Basic information: Name, Description, Category and Classification of the data.
- Custody: Organization it 'belongs' to and Point of Contact (POC).
- Storage information: data storage network(s), type of data, method of storage and data size.
- Relations: Production, Dissemination and Consuming activities (systems) that use, produce or host the data.

To view all the information available on a given data set, one can either go into that specific record or download the Authoritative GOLD copy spreadsheet and see it all in one view. A data dictionary is also available.

The data inventory is continuously updated and new functionality will be added over time. Every migrating activity must provide this information to allow for a successful migration. Data will not be stored in the cloud unless the CDO and/or DC sign off. Continued effort is required by NGA data stewards to ensure the accuracy and completeness of the Data Inventory.

2.0 DATA GUIDANCE

2.1 General Data Guidance

***Guidance:** Data will be managed in accordance with NGA data management, retention and records management policies.*

Data Stewards (DSs) and DCs will set lifecycle rules on their ODS data repositories. Those lifecycle rules will follow National Archives and Records Administration (NARA) guidance and will indicate when data/files/objects can be moved to a near-line archive, and when they can be moved off-line. Electronic records management metadata tags will be applied during conditioning to allow for effective records management.

NGA data management policies shall be followed, as documented in the NGA Data Management Guidance and Unified Data Strategy.⁷

***Guidance:** Data services built for the cloud shall be delivered suitable for use as enterprise services, whenever applicable.*

Conditioning, data load, data access and other data services for applications built for the cloud that may possibly have general (i.e., common) utility to the community are to be delivered so as to be suitable for use as enterprise services. These common data services shall meet standards for web and SOAP services and be documented in the Service Registry.

⁶ [Cloud Migration Data Dictionary](#)

⁷ [NGA Data Management Guidance for EDM](#), 2016; and, [NGA Unified Data Strategy, v2.0](#), 2014.

Guidance: *Data at Rest (DAR) and Data in Transit (DIT) will be encrypted.*

Data at rest (DAR) and data in transit (DIT) across and between environments will be encrypted. NGA follows [DoD's Cloud Computing Security Requirements Guide \(SRG\) v1r2](#), [NIST 800-53r4](#), and [CNSSI 1253](#), all of which provide guidance for DIT/DAR. An Analysis of Alternatives (AoA) has been conducted to evaluate encryption and key management solutions, which will be identified through the NGA acquisition process. Annex I: Cloud Security includes encryption details.

- Data encryption for Unclassified Commercial Cloud (U-AWS). The U-AWS will store, process and transmit both Public Releasable Information and Controlled Unclassified Information (CUI) data (i.e., LIMDIS). The U-AWS will be accredited to process up to Impact Level 4 data. The DoD SRG v1r2 requires that: 1) for all Information Impact Levels, encrypt all DAR, and 2) for CUI data, use a [FIPS 140-2](#) (minimally Level-1) encryption solution.
- Data encryption SC2S and C2S. The Secret Commercial Cloud Services (SC2S) and C2S encryption for DIT/DAR will be accredited in accordance with [ICD 503](#). An objective of the aforementioned AoA is to implement enterprise encryption across U-AWS, SC2S and C2S.

2.2 Mission Data Guidance

Guidance: *Mission data will belong to the NSG community.*

Data does not belong to a specific group or program, but to the National System for Geospatial Intelligence (NSG) community as a whole. Data will be shared unless restricted by access or policy. Whenever applicable, data must be accessible to the entire community including unintended but authorized users. Data access will be determined by the access associated with data itself and the roles and permissions of those requesting access to the data, following NGA releaseability policies.

The DS role does include marking and otherwise associating access and interpretation rules to the data, but a data steward should not support actions that would restrict access by an authorized user with a *need to know*, even if the user or use was unanticipated.

Guidance: *Metadata tagging minimum standards will be enforced.*

Conditioning and validation processes will be part of the workflow; items that fail (e.g., have insufficient handling information or not-discoverable) are so indicated by status metadata or are routed to separate areas (i.e., “dirty” buckets). The minimum standard NGA will follow is dictated by the IC Enterprise Data Header (EDH) coupled with applicable NSG Metadata Foundation (NMF) core elements. Additional fields to handle records management are also mandated but are not necessarily considered for minimum access and discovery capabilities.

Trusted activities, such as those designated for the Activity Data Manager, will have direct access to the data whether it is clean or dirty. Access to “dirty” data is necessary to: 1) get the data conditioned, and 2) enable legacy programs to continue operating while conditioning is taking place. Data must be conditioned before it can be discovered.

Metadata tagging, IAM, and business rules define relationships between data attributes and user credentials. Policy Enforcement Points (PEPs) negotiate access and releaseability. Both enable a strengthened data security posture.

Guidance: *Mission data access shall be permitted in accordance with responsibility to provide principles while enforcing all necessary access controls.*

Mission data access is to be achieved to the extent possible using automated means matching attributes and groups detailing data restrictions and user entitlements. Specifically, entitlement attributes such as, clearance and special access permission group or role membership or non-membership must be explicitly asserted and tested to ensure that:

- Data is tagged with metadata that have classification, dissemination controls, and access group information with associated entitlements policies.
- User identities are tagged with metadata that has clearance, role, and group membership information with access and dissemination consequences.
- Methods, rules and services exist to automatically test data access criteria against access attempt credentials, in accordance with policy.

Access groups and roles can be created at any broad or fine level of detail, by category, by inclusion, by exclusion, or by specific itemization (e.g., listing who, what, and when).

Guidance: *Mission data shall be approved for migration with the CDO, DC and DS.*

Before any set of mission data can be migrated to C2S, the CDO, the mission DC and DS must approve (by electronic signature) the dataset's migration. If no one claims ownership or provides stewardship of a set of mission data, that data will not be migrated. If the mission data are deemed no longer relevant or active, then the DC or DS (or CDO) may choose to not migrate the data. Then, data management would fall under NARA records management rules including potential disposal or archiving. The approval is only needed for the initial migration, not new updates or additions to the existing data set.

Guidance: *Mission data will be migrated to an OpenDataStore (ODS).*

C2S/ AWS shall be the default storage location of all NGA mission data. If IC partners request NGA data on the IC GovCloud for analytical purposes, a copy of the data requested can be made available at the requested location. IC GovCloud usage may be allowed for other reasons on a case-by-case basis.

Guidance: *Mission data will be stored in an OpenDataStore account.*

All mission data shall be stored in the ODS under the NGA mission content account in C2S, or other accounts managed by the CDO. This underscores enterprise data ownership, and helps guarantee uniform accessibility while maintaining necessary stewardship and access controls. Within the ODS, numerous different databases, archives, S3 buckets, and other data stores can be created, and managed, with the appropriate permissions and roles.

A system, program, group or individual may still have a separate C2S account with any permissions necessary for special purposes, as long as the mission data is in ODS and available to the community.

Guidance: *Mission data will be cataloged for discovery and exposed via services.*

Activity owners and designers should work to maximize discovery and the potential for reuse. Specifically, they shall maintain and catalog metadata that describe mission data, such as the EDH and the information contents of the Trusted Data Format (TDF), including provenance and access information. Whenever practical, discovery data and content data are to be not directly accessible; but, only through services which control access and track data use. Policy Enforcement Points

(PEP) enforce user access to all data stores via a service or view with no direct access. Application data queries will be encapsulated as services exposed to the community and published in the service registry.

Guidance: *Mission data will be managed with integrated security and access control.*

The cloud implements multiple layers of defense, or defense-in-depth, to protect against unauthorized access and the introduction of vulnerabilities (e.g. malware) into the NGA environment. The cloud infrastructure is protected by packet analysis, data loss prevention, auditing, privileged access management, and encryption for DAR and DIT. In addition, various other security services are employed to provide extra layers of protection. DSs and DCs are to leverage Public Key Infrastructure (PKI), IdAM, role-based access control (RBAC) and PEPs to maintain secure access to data and metadata.

2.3 Activity Data Guidance

Guidance: *Activity owners and designers shall determine the most suitable activity data storage type and location, with the exception of logs.*

Security and audit files will be accessible by qualified security administrators only. See Annex I: Cloud Security and the Design White Paper on C2S Storage Service Options⁸ for additional information. Redmine and Gitlab are the approved CM/registry applications for DevOps activities. GitLab will be used to store development code while ODS is suitable for storing NGA content, source, and mission data.

2.4 Analytical Data Guidance

Guidance: *Analytical data will be stored as determined by owners of the enterprise analytical services and exploitation services.*

Best practices suggest that most analytic data should be dynamically compiled in cache, if practical, or in temporary data structures and automatically deleted at the conclusion of the analysis. Certain analytic situations may require other data storage and purge procedures, at the discretion of activity owner.

2.5 Corporate Data Guidance

Guidance: *Corporate data will be access-controlled.*

The NGA ODS account will have protections in place to address security controls for PII, financial, Human Resource, and other sensitive non-mission data types. Security measures such as RBAC, PEPs, and business rules will be employed to control access to both discovery metadata and data content.

Guidance: *Corporate data will be migrated to an ODS.*

The CDO has specified that ODS C2S will be the migration destination and storage location of all NGA corporate data.

Guidance: *Corporate data will be stored in the ODS.*

⁸ [C2S Storage Service Options, 2018.](#)

All corporate data will be stored under an ODS account. This underscores enterprise data ownership, providing uniform accessibility while maintaining necessary stewardship and access controls. A system, program, group or individual may still have separate accounts with any permissions necessary. Within the ODS, numerous different databases, archives, and other data stores can be set up, and managed, with the appropriate permissions and roles.

Guidance: *Corporate data will be cataloged for discovery and exposed via services.*

Corporate DCs shall publish discovery metadata in a corporate metadata repository distinct from the mission metadata repository. Whenever practical, application data queries will be encapsulated as services exposed to the community and published in the service registry

Guidance: *Corporate data will be managed with integrated security and access controls.*

The cloud implements multiple layers of defense, or defense-in-depth, to protect against unauthorized access and introduction of vulnerabilities (e.g. malware) into the NGA environment. The cloud infrastructure is protected by packet analysis, data loss prevention, auditing, privileged access management, and encryption for DAR and DIT. In addition, various other security services are employed to provide extra layers of protection. DSs and DCs will leverage PKI, IdAM, RBAC and PEPs to maintain access to data and metadata.

2.6 Organizational Data Guidance

Guidance: *Organizational data currently on a shared drive and/or SharePoint will be migrated to the equivalent NGA enterprise location in the cloud.*

No specific, individual organizational actions are needed at this time; DSs may relocate this data at their discretion.

3.0 DATA MIGRATION

3.1 Background

The NGA mandate⁹ is to migrate all content and activities to the cloud. The key migration objectives are:

- Do not break mission capabilities,
- Get off the current infrastructure,
- Obtain secure, efficient, intelligence integration.

As shown in Figure 4, migration is dependent on both the data and activity migration. Mission migration is not complete until the applications, services, and data are usable together.

⁹ [Memo: NGA Cloud Guidance, U-2016-00069, 10 February 2016](#)

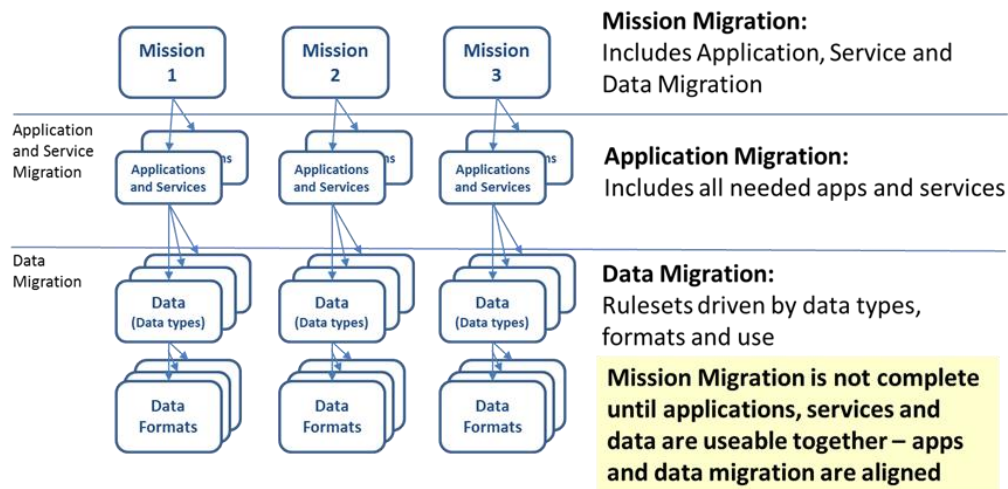


Figure 4: Cloud Migration Dependences

Mission data migration ahead of the activities helps to accelerate customer consumption, activity test readiness and interim operations.

3.2 Data Migration Guidance

Guidance: Program Managers and developers shall support the migration effort by implementing the Migration Staging Strategy.

The Migration Staging Strategy follows:

- Provision staging S3 buckets by source, data type and classification.
- Provision additional buckets as needed (e.g., Special Access Program (SAP) data).
- Provision a “Manual Review” dirty bucket per migrating system for all source data without appropriately tagged metadata. Properly tagged legacy data may skip the staging dirty bucket.
- No end user access to the “Manual Review” dirty bucket. DSs must have access to remediate the content and workload managers must have oversight permissions.
- Do not mix data in the staging architecture that is not mixed today.
- Migrate non-file-based data by following database criteria guidance.

Guidance: ‘Interim’ operational access will be granted if content is not in desired organizational structure.

Guidance: Program Managers and developers shall incorporate resiliency requirements for production storage in the design (e.g., multiple availability zones) for operational data.

Guidance: Program Managers and developers shall implement persistent data storage (e.g., S3 Bucket Strategy).

The S3 Bucket Strategy follows:

- All persistent storage for all corporate or mission file-based data will be in S3, with archives using Glacier capabilities.

- Use standard enterprise S3 buckets, as predefined.
- Content-IDs are assigned to data at the logical dataset level.
- Virtual folders are set up for each ContentID within an S3 bucket, managed by activity's data manager.
- Lifecycle/data retention rules are set at the ContentID level.
- Permission for direct S3 bucket access by trusted activities are set at the ContentID level with:
 - Read only access by default.
 - Write/delete access (activity data manager) requiring ContentID approval (i.e., via DC).
 - Trusted activities, if not using the enterprise PEP, responsible for ensuring appropriate access by users.
 - Content following S3 bucket naming standards, presented in [Appendix H](#).
- S3 buckets used will be the standardized S3 buckets set up under the ODS.
- S3 bucket naming convention will be:
 - \\<S3 Bucket Name>\<Content-ID>\<Managing Activity>
- Special access permissions can be granted at any level of a virtual folder structure.
 - AWS enables wildcards within the string for access policies.
- Lifecycle policies will normally be set at the Content-ID level per DS decisions.
 - AWS only enable wildcards at the end of the string for lifecycle policies.
- ATO of the ODS services, including the Metadata Catalog, are a DC responsibility.

3.2.1 Data to the Cloud Process

The process of getting data to the cloud is outlined by the steps to: identify content, approve requests and create/implement ODS change requests that will apply to most datasets. This process applies to moving data to the ODS, establishing storage in the ODS with the appropriate accesses, and creating data feeds. Each data set may have unique issues and considerations. ODS cloud migration preconditions are:

1. A Data Custodian must be identified,
2. Data must be registered in the data inventory such that a ContentID is associated with the data, and
3. Essential metadata (a minimum set) must be populated for the particular data set.

This process is the same for both file/object based data and database data with a few exceptions: 1) database data will not be posted to a dirty bucket, but instead to its designated database structure, and 2) while the ISP and DSP will move database data, as they move file/object data; it will be up to the ASP to ETL data into the cloud's database.

3.2.1.1 Phase 1: Identify Content

The identity phase begins when a party (CDO, DC, DS, or PM) identifies data to be migrated to or established in the ODS.

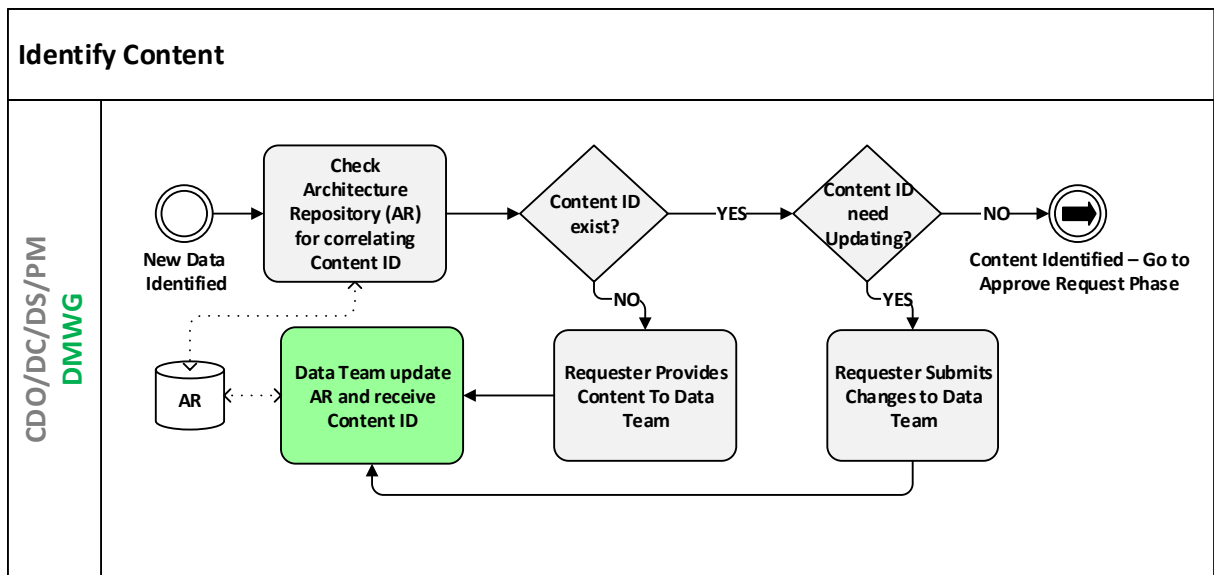


Figure 5: Phase 1 - Identify Content (assign Content-ID)

As shown in Figure 5, all parties must check the ATLAS's AR (i.e., [Data Inventory](#)) to see if their content aligns with any content already cataloged.

- If their content is not found the initiating party shall contact the Data Team to generate a Content-ID for the new activity data.
- If the requester finds their Content-ID(s) but any of the metadata is incorrect, the requester submits changes to the Data Team.

It is important to note that at this point there must be one DC identified for the Content-ID. Data Custodians can own multiple Content-IDs but only one DC can be identified per Content-ID.

3.2.1.2 Phase 2: Approve Request

The approval phase begins when the CDO/DC/DS/PMs compile and submit their particular Content-ID(s) for approval (i.e., migration, data feed set up, create storage and access for repository space), as outlined in Figure 6.

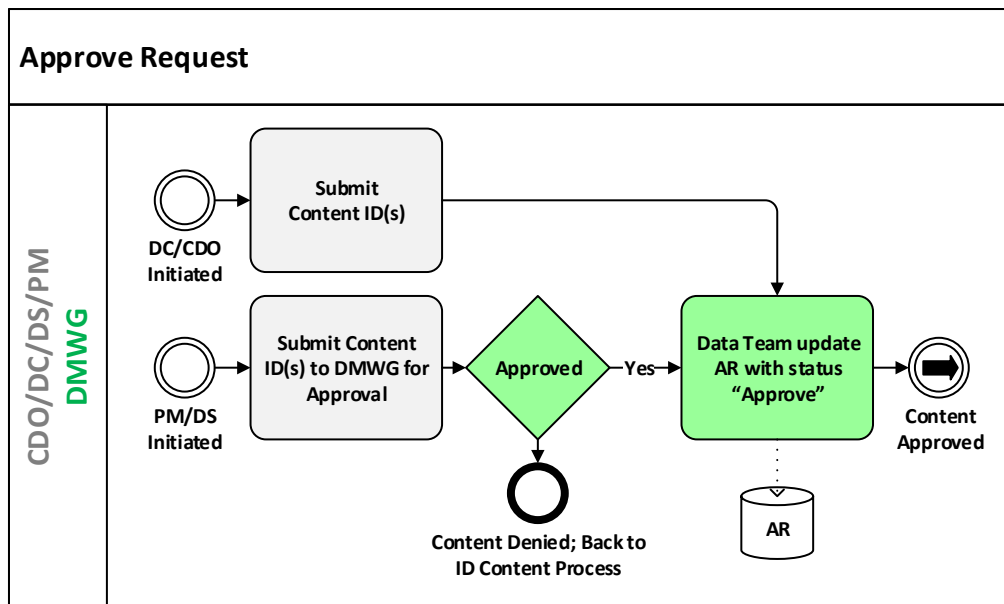


Figure 6: Phase 2 – Approve Request

The DC for the particular Content-ID(s), as a representative(s) of the Data Management Working Group (DMWG), reviews approval requests.

- Content-IDs are checked and the information is verified by the DC and Data Team. Additional data, if missing, may need to be collected by the DC.
- Whether the content is approved or denied, the repository is updated by the Data Team to reflect the status of the data (i.e., migration, data feed set up, create storage and access for repository space).

Note: If the CDO or the DC submits the data for approval, the DMWG is bypassed and the repository is updated by the Data Team to reflect approval and any additional information, if needed.

3.2.1.3 Phase 3: Create/Implement Cloud Change Request

The implementation phase begins with all submitting parties (i.e., CDO, DC, DS, PM or representative) submitting to the ODS, a request to move, create storage and access for, or create data feed for a set of one or more Content-ID(s) that was already approved out of Phase 2. The change request (CRQ) creation and implementation process is outlined in Figure 7.

The ODS verifies that this request is coming from the appropriate managing activity, DC/DS, or the CDO for the Content-ID, else the request is rejected. A change request is generated then TEMs are scheduled for the ISP, DSP, and ASP. After the details are worked out the request is granted (i.e., data moved, storage created, data feed set up), the AR updated, and tools are pointed to data's ODS location.

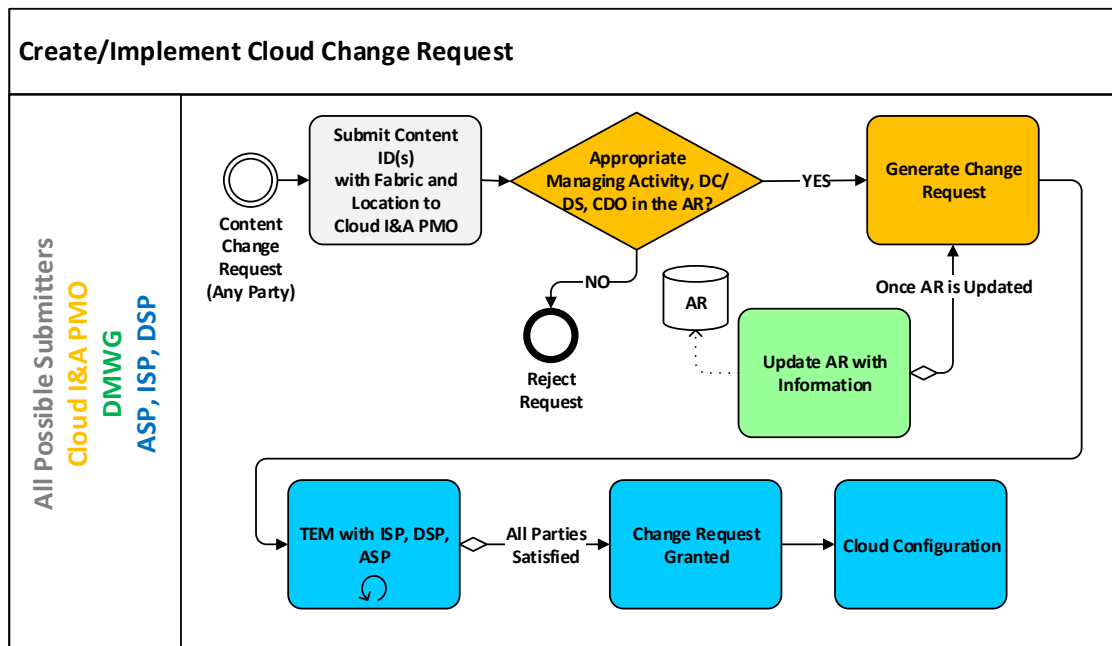


Figure 7: Phase 3 – Create/Implement Cloud Change Request

3.2.2 Conditioning for File/Object based Data

All cloud data needs minimum metadata (specified in [Appendix F](#), including GUIDE ID) to be discoverable and accessible by the enterprise. The process of defining and adding this metadata is called “Conditioning”. If the minimum discovery and access metadata is not immediately available when the data is moved, the data is placed in a logical ‘Dirty Bucket’ and further conditioning is scheduled. The following provides a high level overview of the steps in the data movement process.

1. Originating Activity Data Conditioning. In many cases, only the originating activity will have the information and tools needed to condition the data. In these cases, the activity should submit the metadata along with the actual data files, as part of the data movement process. Once the initial migration is completed, to support ongoing operations, the requisite data conditioning procedures and processing information need to be incorporated into an Enterprise Conditioner Service.
2. Enterprise Conditioner. In some cases, there is an Enterprise Conditioner already created for the file type of the data you are migrating. In this case, the DS would be responsible for running the Enterprise Conditioner after the data is migrated.

If there is no Enterprise Conditioner, and the required metadata cannot be provided by the Originating Activity, then the DC needs to approach the CDO via the DMWG with the request to create a conditioning tool. The CDO office will work with the DC on prioritization of conditioning tools and strategies to create the tools in alignment with mission needs. In some cases, information needed to condition data is not stored in any digital way – the DS will be responsible for transferring that corporate knowledge to the metadata catalog.

3.2.3 Conditioning for Databases

Conditioning for databases will be the registration of all the web services associated with that database in the enterprise service registry and populating any mandatory metadata there. No changes will be made to the database internal schemas, so as to not break legacy functionality.

3.2.4 Production Components

The production architecture will be composed of catalogs and registries for all domains, file-based content where the content is persisted on the originating domain, and non-file-based content that is either in an interim or final operational state. Figure 8 identifies the different data components. The definitions of each are found in [Appendix A](#).

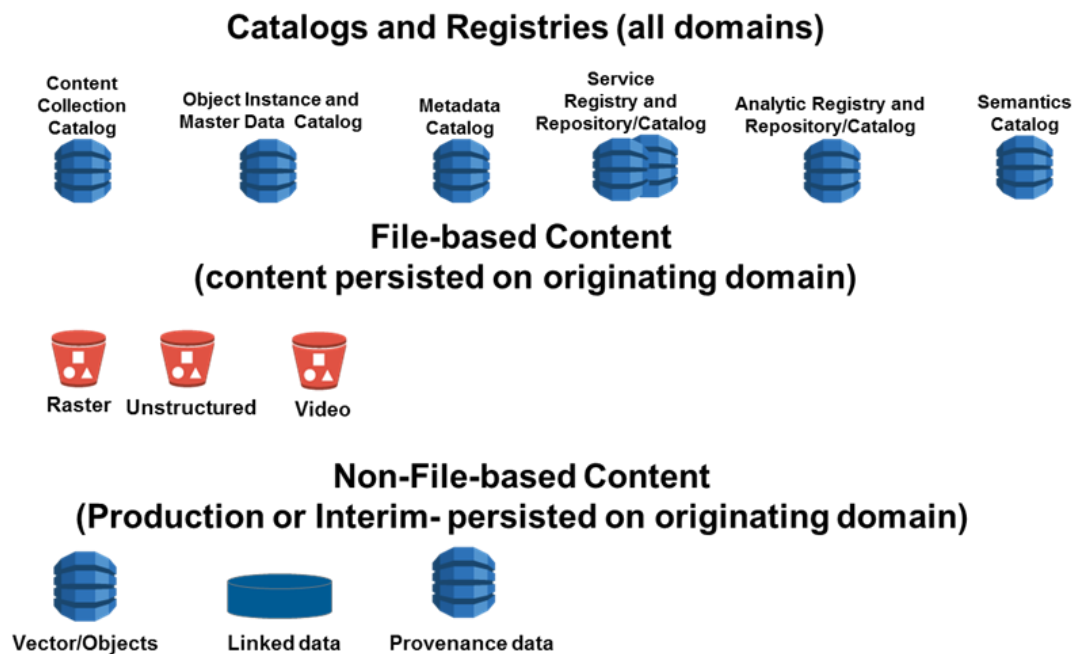


Figure 8: Production Architecture Components

3.2.5 Cloud Migration Lessons Learned

The feedback loop can be a valuable aspect of migration management. Cloud migrations' Lessons Learned are captured in the NGA Lessons Learned Portal. The Lessons Learned Portal is a way to review migration issues and avoid the pitfalls experienced by others in their migration efforts. It is also an opportunity for groups to provide feedback to the management teams and reflect implementation best practices.

3.3 Metadata Management

The NGA's mission data has historically been difficult to manage through the ingest to dissemination production lifecycle. Data access, discovery, and retrieval have been problematic; partly because of inconsistent metadata conditioning.

For managing cloud data, all NGA mission data will be stored in C2S; and, NGA will push metadata to IC GovCloud, as appropriate. +Mission data/files will be submitted to a service that will condition the data. Once conditioned, mission data and its metadata will be stored separately with data going to an Object Store and metadata going to a Metadata Catalog.

Database-based data will not undergo direct conditioning. Instead, the service that manages that database-based data will be registered in the NGA Enterprise Service Registry that has metadata that allows for discovery.

Data conditioning requires two things: 1) knowing the data, and 2) a consistent way of describing data that will be understood by human and machine consumers. To meet those requirements NGA will enforce DNI and NGA metadata standards for C2S data conditioning. Complying with the TDF, EDH, and NMF standards will ensure that not only NGA data is discoverable, but that our mission partners will be able to discover and use that mission data.

3.3.1 Challenge #1: Increasing Discoverability of Mission Data

Historically, NGA has found making its mission data discoverable to the uninitiated user to be challenging. The consumers needed to know where to find their required data and products, typically by navigating a website to arrive at a tool that allowed them to download the needed data based on additional search criteria.

3.3.2 Challenge #2: Securely Sharing Information with External Partners

Legacy systems have implemented stove-piped architectures with their own proprietary access controls to protect mission data. This has hindered the sharing of information between IC agencies as well as with non-IC agencies. If a non-US IC organization required access to data, and they were not eligible for access to all the information on a given network, a separate site had to be created for them to retrieve their data because it could not be securely filtered. Posting data this way led to currency issues and duplicative storage.

IC ITE introduces enterprise security services to eliminate these proprietary access controls and provide a uniform method for accessing mission data from any agency. These services secure data at the object level. Redaction services used to filter content within an object are not included. NGA will supplement IC ITE's core security services with these capabilities. Access control decisions at the object level will be governed by DNI's security specifications (e.g., EDH). Objects requiring additional content inspection and filtering will be governed by NGA-specific metadata.

3.3.3 Tagging Mission data to Overcome Challenges

Tagging data with metadata specified by both the DNI and NGA can help improve discoverability and sharing of mission data. By leveraging the cloud, enabling enterprise access controls, and enhancing and standardizing metadata, the following opportunities can be realized:

- Consistently metadata tagging data throughout the NGA production process from ingest to dissemination.

- Enable search of all NGA data through consistent metadata for NGA internal and external users.
- Automated information release becomes possible.
- NGA can inventory its own data and assess its overall enterprise value for enhanced sharing opportunities or opportunities to retire obsolete data based on usage statistics.
- Automated data management at an enterprise level becomes real. Appropriate metadata tagging provides for automated tools to enable decision making, therefore reducing the manual process of assessing data.
- Enabling enterprise level automation of high-to-low transfers.
- Consolidation of disparate security services within the NGA enterprise.
- Solutions can scale for user and data access management.

The NGA stood up the Office of the CDO to address data issues, among them metadata and discovery. Managing data as a true shared asset will greatly improve and indeed revolutionize the way NGA fulfills its mission.

3.3.4 Metadata Management Guidance

The CDO's metadata management guidance includes correcting the metadata/handling/discovery issues in the cloud. There are two broad categories of mission data: "file/object" and "database" based data.

Guidance: *Metadata management for file/object-based data shall include:*

- *Conditioned (metadata tagged) mission data objects posted to the ODS.*

In order to ensure full discovery, access, and retrieval, all NGA mission data files/objects will be conditioned. This conditioning allows for uniform application of metadata to all NGA mission data. Uniform metadata tagging provides for more autonomous data management procedures.

- *NGA mission data identity and discovery metadata posted in the NGA Enterprise Catalog (containing one index record for each object).*

Metadata conditioning will resolve NGA data discovery shortfalls by attaching a well-defined set of attributes to all mission data files/data objects. These attributes, to include security markings, can then be cataloged and searched for every mission data object in the NGA enterprise. As a side benefit, this metadata catalog effectively creates an enterprise inventory for all NGA file-based mission data.

- *Use of Globally Unique ID (GUIDE) to uniquely tag each object and as the means to associate the object with its index entry in the Enterprise Catalog.*

The NGA will use Global Unique Identifier for Everything (GUIDE) in accordance with DNI Standards for Intelligence Community Identifier. Application of the prefix defined in [GUIDE](#) will be required but registration with the community GUIDE is encouraged but not required. Note that for technical reasons, GUIDE is not recommended for use as an indexed key-ID field in some storage solutions such as S3.

- *Application of DNI directives on using TDF and EDH metadata standards for object conditioning, with:*

- *Additional NGA Mandatory Conditioning Fields tags above and beyond DNI mandatory fields to be applied as part of mission assertions.*
- *TDF and EDH metadata in the NGA Enterprise Catalog to be used instead of using the TDF as an object wrapper for persistent storage.*

Conditioning is physically implemented in one or two ways. The first is referential, in which metadata are captured separately in a metadata catalog, linked to the object via Uniform Resource Identifiers (URI). The second is embedded, by which the metadata are included in a wrapper or envelope in which the actual object can be found, or directly in the object information. Multiple technical approaches are possible. It may also be mentioned that transport metadata are a different subject – TDF is an example of a transport metadata encapsulation protocol that can also be used for persistent storage. NGA will be doing a referential implementation in a metadata catalog, not wrapping the files.

If every mission data object is conditioned for ingest into the ODS, then this includes all incoming sources. When every piece of incoming source is conditioned then it is possible to inherit the metadata from source to all products and datasets subsequently generated, thus preserving pedigree and lineage.

With everything conditioned with the appropriate metadata it is possible to build automated PEPs to process and release data based purely upon a combination of mission object metadata and consumer permissions as granted by the IdAM system. Some attributes identified in the IdAM System correlate to the attributes used in the EDH. Doing this reduces the need for excessively redundant storage and manual releaseability processing.

The effectiveness of an enterprise IdAM system and data access management depends on the consistency of tagged data (i.e., compliance with standards).

The accuracy, consistency and robustness of data tagging is critical to support enterprise-based releaseability and access control functions. The required enhancements to the security markings and metadata tags need to be identified to enable the programmatic enforcement of releaseability rules and policies.

3.3.4.1 NGA Metadata Catalog

NGA will establish its own metadata catalog to account for additional metadata beyond the IC minimums stored within the IC GovCloud. Since NGA will be storing all of its authoritative data in C2S, it will stand up its catalog within C2S.

All ingested file/object mission data will be submitted to the NGA Conditioning Service. The NGA Conditioning Service will enable proper metadata tagging and then build a reference to the data. Note that during an initial migration, conditioning may be performed by the originating activity. The data will then be sent to an Object Store(s) while the metadata will be sent to a Metadata Catalog, as shown in Figure 9. Handling data this way allows for uniform search and retrieval protocol, as well as provide a central location for metadata standards to be enforced. Note that Figure 9 is conceptual and is not meant to represent an actual design or implementation.

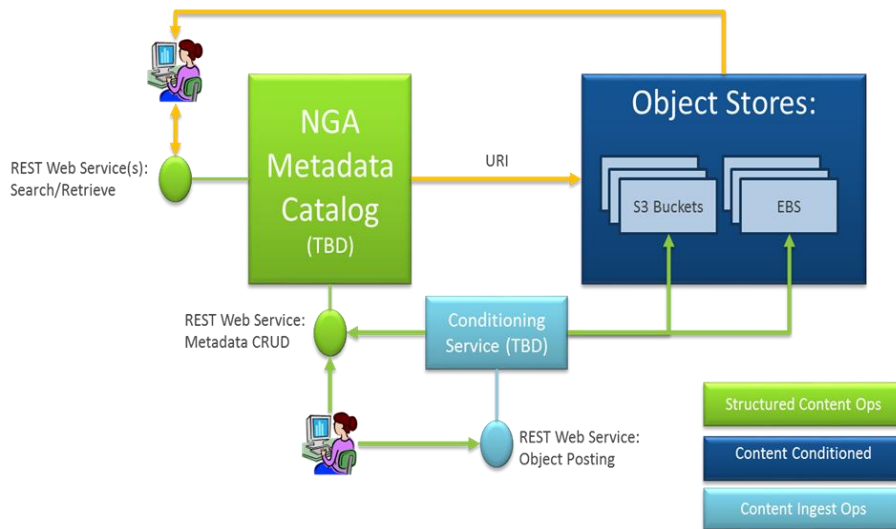


Figure 9: Concept: Metadata and Data in C2S

While the comprehensive NGA Metadata Catalog will reside within C2S this does not mean that NGA will not populate the IC GovCloud metadata repositories. NGA will work with the IC to determine what metadata will be parsed and pushed from the NGA repository to the IC GovCloud. Figure 10 illustrates this concept.

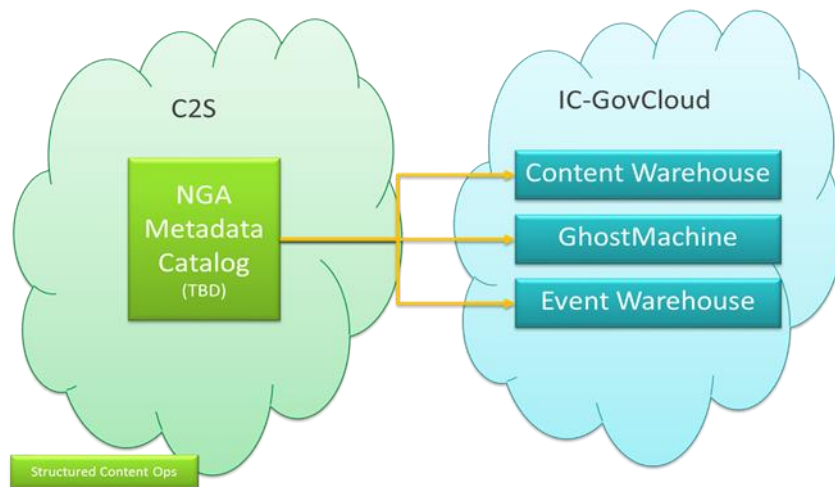


Figure 10: Concept: NGA to IC GovCloud Exchange

There will be an NGA Metadata Catalog substantiated in the NGA unclassified cloud hosted by AWS Commercial.

3.3.4.2 EDH and TDF Use Criteria

Intelligence Community Directive (ICD) 501 authorizes the DNI to produce and mandate metadata standards for the IC to adhere to in the cloud. Following that, the DNI has published guidelines and specifications for the use of EDH and TDF to condition and standardize mission data for IC use.

The NGA has traditionally followed the NMF to specify mandatory NGA metadata. The CDO has mandated the use of NMF mandatory content in concert with the EDH and TDF in the NGA

Metadata Catalog. To comply with those mandates NGA has gathered information to aid content holders:

- NMF has mandatory fields specified in NMF Part 1. (See [Appendix C.1](#)).
- A and S organizations have specified fields in the NMF that they jointly consider mandatory (See [Appendix C.2](#)).
- X organization has submitted a list of discovery metadata that they require to the CDO. (See [Appendix C.3](#))

The question is “How do we reconcile the mandate to use EDH and TDF with the mandatory metadata from the NMF?” Apply the following steps:

1. Identify mandatory fields in TDF and EDH. (See [Appendix E.2](#) and [Appendix E.3](#), respectively)
2. Identify the entire list of NGA NMF-based mandatory fields. (See [Appendix F](#))
3. Cross reference the NGA NMF mandatory fields with the TDF and EDH. (See [Appendix D](#))
4. Add Mission Assertions to EDH and TDF to cover NGA NMF mandatory fields that didn’t map and create a NGA profile of the TDF and EDH. (See [TDF’s XML portion](#))

3.3.4.3 EDH

The DNI specification for metadata calls for use of EDH. It establishes a standard for universally and consistently applying a small set of metadata tags to any piece of data so that it can be identified, protected, tracked and handled throughout its life cycle within the enterprise, from creation to deletion. The EDH includes most Access Rights and Handling (ARH) attributes associated with an object. The ARH attributes must be taken into consideration for discovery, dissemination, and access authorization services. Figure 11 is a graphical depiction of the EDH. EDH attributes and details are presented in Appendix E.3.

NGA will follow the DNI specifications for all the boxes represented in Figure 11. The delivery mechanism is supported by the TDF.



Figure 11: EDH Graphic

3.3.4.4 TDF

NGA will follow the DNI specification of TDF metadata content. TDF exchange standard features a general container structure that encapsulates data objects and any assertions made against them. The TDF is an extensible, self-describing, self-protecting container/wrapper for assigning secure attribute assertions to any type of data. Figure 12 is a depiction of how the DNI conceptualizes the TDF structure. TDF attributes and details are presented in [Appendix E.2](#).

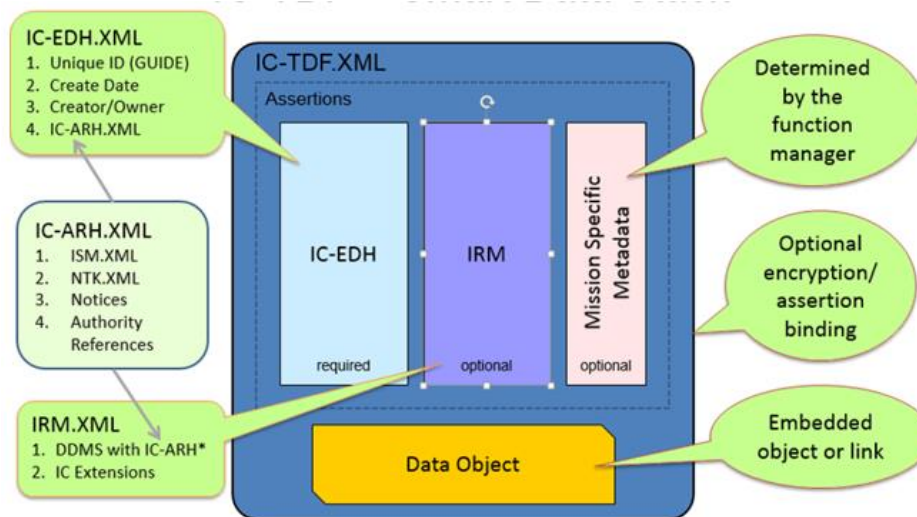


Figure 12: TDF Graphic

The TDF breaks down into two main parts. The assertions (metadata describing the Data Object) and the actual Data Object (mission data). Using the TDF allows NGA mission data to become Trusted Data Objects (TDO). Under that format, additional assertions beyond EDH are made.

3.3.4.5 TDO

A TDO is a wrapper containing a single payload. Each TDO requires at least one handling assertion at the package or metadata level (never encrypted), optionally a payload handling assertion, optionally several mission assertions, and payload. Handling assertions explain who can use the content and to what ends. Typically, it is the handling assertions described as the EDH that are sent to the IC GovCloud.

- Payloads are the actual mission content/ data object. It can be a string, eXtensible Markup Language (XML), or a reference to an external object.
- Mission assertions are only mandatory for the NGA Metadata Catalog within C2S. There are DNI standards to inform content providers (See [Appendix B: References](#)).

3.3.4.6 Assertions

The NGA mandates other assertions on top of what the EDH provides. This collective group of assertions are known as mission assertions. These assertions allow for a more robust set of access, discovery and retrieval attribution. The NGA/NSG mission assertions are specified in Appendix G.

3.3.4.7 NGA Web Service Registry

Guidance: Metadata management for database-based data shall include services that expose database data registered in the NGA Enterprise Service Registry.

NGA will take a different approach to database-based data. Instead of tagging every record, NGA requires that the web service that exposes that data to be registered in the NGA Web Service Registry with appropriate metadata in the registry to make the service discoverable. Consumers can then search the NGA Web Service Registry for web services that expose their desired content. Figure 13 is a graphical depiction of the NGA Web Service Registry architecture.

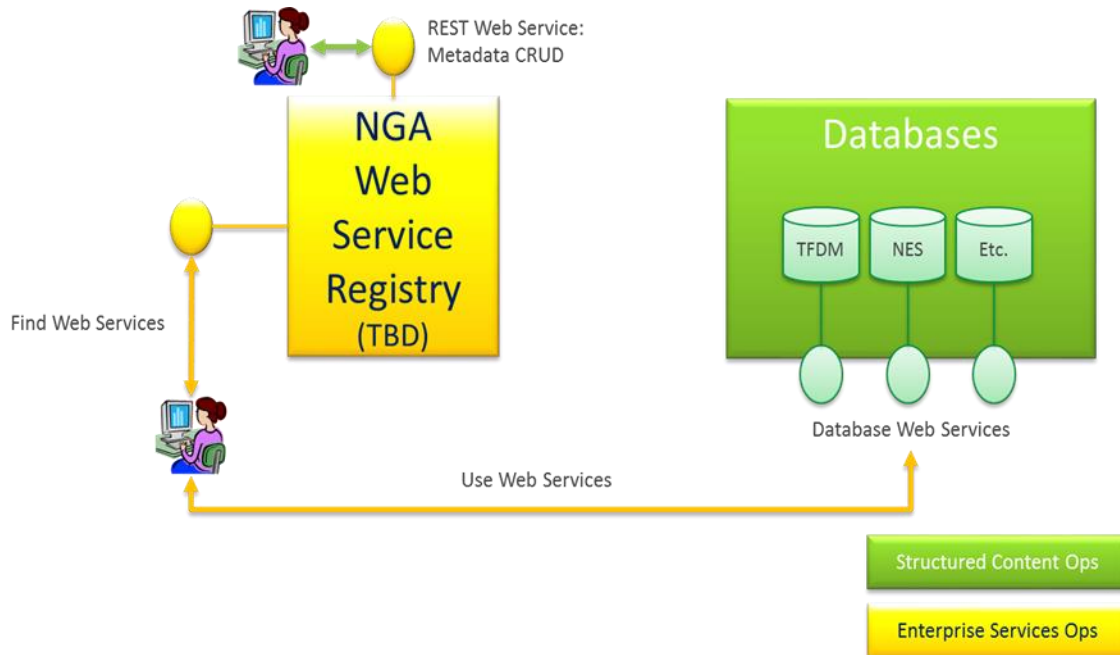


Figure 13: NGA Database-Based Discovery via Web Service Tagging

4.0 DATA ACCESS VISION

In the past, data access has been system-specific and typically enforced by access/login protocols, and often not as fine-grained as desired or necessary. With IC ITE, data is managed independent of originating systems; and, access is controlled by policies referencing attributes of the data and the user. In alignment with DNI directives, the data access architecture for mission data has the following features:

- An automated way to discover and access data. All data is metadata tagged, personnel identities or roles are associated with accessibility attributes, and automated rules and processes will use these tags to automatically enforce accessibility and dissemination policy.
- Responsibility to provide is achieved. The default stance on data is that it is shareable, within the classification and dissemination constraints as asserted by the security markings. However, *responsibility to provide* does not negate our responsibility to protect sources and methods.

- Person entities gain access with the “appropriate security clearance and an assigned mission need”. Entities are identified by security attributes and roles are able to access data with corresponding attributes and roles. These assignments and entitlements are how “need to know” is expressed and implemented. No other “need to know” check is performed – it is implicit in the request.

4.1 Responsibility to Provide

Need to Know (NTK) has been modified to fulfill the *Responsibility to Provide* concept. [ICPM 2007-200-2, “Preparing Intelligence to meet the Intelligence Community’s Responsibility to Provide”](#), and [ICPM 2007-500-3 “Intelligence Information Sharing”](#) directed the information shift from data enclaves to data sharing among all intelligence elements, rescinding some directives that had prevented sharing. [ICD 501, “Discovery and Dissemination or Retrieval of Information Within the Intelligence Community”](#), states the following:

“Responsibility to Provide:

a. IC elements shall fulfill their “responsibility to provide” by making all intelligence and intelligence-related information (hereinafter referred to as “information”) that IC elements are authorized to acquire, collect, hold, or obtain (hereinafter referred to as “information collected”) or analysis an IC element is authorized to produce discoverable by automated means by “authorized IC personnel,” in accordance with Section D, unless otherwise exempt in accordance with Intelligence Community Policy Guidance (ICPG) 501.1, Exemption of Information from Discovery. Authorized IC personnel are individuals identified by their element head and who have an appropriate security clearance and an assigned mission need for information collected or analysis produced. “Discovery,” as defined in Appendix A, is the act of obtaining knowledge of the existence, but not necessarily the content, of information collected or analysis produced by any IC element.

b. “Stewards,” as defined in Appendix A, shall fulfill their “responsibility to provide” by making all information collected and analysis produced by an IC element available for discovery by automated means by authorized IC personnel [emphasis added], unless otherwise determined by the DNI; by making as much information as possible available for automated retrieval upon discovery; and by presuming that authorized IC personnel who request information discovered possess a “need to know,” in accordance with Section F.”

The IC is directed to presume that anyone authorized for access has a need to know. Marking all intelligence data/products with the proper security markings per DNI standards, such as the IC Information Security Marking (IC ISM) metadata is one prerequisite. Others include maintaining entitlements categories and access roles associated with authorized personnel, and policy/rule-based automated enforcement mechanisms.

Note that Personally Identifiable Information (PII) and Personal Health Information (PHI) are NOT covered by “*Responsibility to Provide*”. Specific, explicit and demonstrable “need to know” applies without exception.¹⁰

¹⁰ [DoD Directive 5400.11, DoD Privacy Program, 2007.](#)

4.2 Authorized Data Access

The IC is directed to provide a means of automated discovery and retrieval for authorized personnel, by data content and data service without necessarily resorting to the “owning” application. Therefore, access needs to be controlled and granted at the data level, not at the application level, to make Data-as-a-Service work. Individual systems can no longer be the sole gatekeepers to data access. All authorized people and entities must be able to get access to the data without setting up an account with a specific individual system or navigating through one owning application. Systems become the consumers of data instead of the owners of data.

4.3 Data Entitlements

Entitlements are rights and permissions that allow an entity privileges with data. Entitlements management is the set of procedures and technologies used to grant and revoke access rights and privileges to entities in support of authorization determination requirements.

Specific entitlements control action permissions (e.g., allow one to GET a dataset). Entitlements are used by policy-based rules (e.g., Policy Decision Point (PDP)) to execute data access and provisioning decisions based on entity attributes and identifiers. Within a policy context, this represents the combination of:

- Privilege,
- Entity (i.e., person entity (PE) or non-person entity (NPE)) assigned an entitlement,
- Applicable data, and metadata
- Conditions (i.e., environment, time range, geographic range).

All of these are needed to determine accessibility, and the nature of that access. The other needed ingredient is the set of rules and methods that procedurally implement policy.

4.4 Data Privileges

When viewed from a data viewpoint, there are a limited number of things a person can do with data. These fall into three broad categories of privileges: READ, MANAGE and ADMINISTER. The focus of this guidance is on READ and MANAGE.

- **READ** is when the entity has access to the data, but is not able to change the data in the repository. The two specific access privileges for READ are:
 - LIST: This allows for a list/view of the data online. It does not allow for download. This is equated to ICD 501’s discovery.
 - GET: This allows for the download of a copy of the data. This is often equated to ICD 501’s dissemination.
- **MANAGE** is the ability to change the data in some way. Some groups allow analysts to have some or all of the MANAGE privileges, others allocate some, or all, of these privileges to administrators. The specific access entitlements for MANAGE are:
 - INSERT/PUT: The ability to “PUT” a new file in an S3 bucket or to “INSERT” a new record/document in a database.
 - APPEND: The ability to APPEND new information to an existing file/record/document without changing existing information (e.g., analysts append new information to an image record).
 - UPDATE: The ability to change information in an existing file/record/document.

- **DELETE:** The ability to DELETE an existing file/record/document.
- **ADMINISTER.** The list of things an administrator can do is far longer and is often more specific to the actual storage solution. However, some generic examples are the ability to backup and restore data.

4.5 Legacy Data Access

Legacy activities (i.e., owning application) act as the gatekeeper for their own data; the only way to get to the data is via the activity application, via login. Some legacy activities validate the identity of users accessing the application with GeoAxIS. Very few activities do actual authorization inside of their application, with most allowing complete READ access to everything in the system, via login, and write access based on application roles. As such, most of the policy enforcement done by the application is really up front when setting up the user accounts: they validate that the user is allowed to see the data before granting an account, per Figure 14.

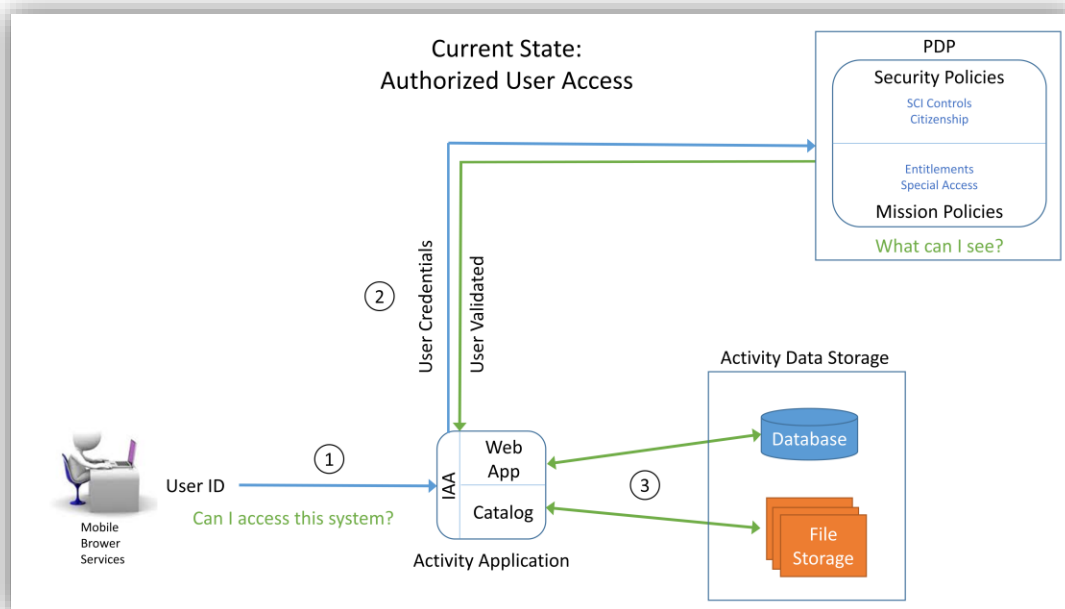


Figure 14: Legacy Data Access

Demonstrating the legacy data access, in Figure 14:

1. A user logs into the activity application with his User ID.
2. The IAA portion of the activity sends that ID off to the PDP that validates whether he is an authorized user.
3. Once validated, he has access, via the application, to the database and file storage that activity might control.

4.6 Legacy Network Access

One of the data access methods practiced at NGA is “if you have access to the network, you have access to the data”, also often referred to as Lowest Common Denominator (LCD) access. This practice led to the setup of redundant sites and storage locations where NGA would duplicate subsets of data that specific mission partners were allowed to have, and place it on their specific site because the mission partners could not be allowed access to the main network without exposing data they were not entitled to. Data duplication and subset database are no longer is considered appropriate. Going forward all data moving to the cloud will be conditioned with the appropriate metadata. Data will be stored once, and accessed in accordance with the IC’s data access standards.

4.7 Automated Data Access

For automated data access, data are tagged with security and access metadata attributes (including but not necessarily only ‘Classification and Control Markings’) specifying restrictions. On-line identities, for PE or NPEs are tagged with authorization attributes, such as in the federated Authoritative Attribute Stores (AAS). Any additional access controls by role membership are also checked. The enterprise PDP compares the data security attributes against the entities’ authorization attributes and automatically will allow or deny access to the data.

The DNI concept of automated access and discovery with the appropriate clearance and assigned mission is served by two basic use cases: 1) Access by authorization attributes, and 2) Access by role.

4.7.1 Access by Authorization Attributes

This type of access control is usually called Attribute Based Access Control (ABAC), where one’s specific attributes allow access to the data.

For an access granted example, the Maritime Tactical Ocean Data 4 (TOD4) product is tagged as CONFIDENTIAL //NOFORN. No other special access or dissemination restrictions apply to the data set, so no role checks are necessary. Mary is a US Citizen with a Secret Clearance. Figure 15 illustrates how her access attributes align. She will be granted access to LIST or GET TOD4.

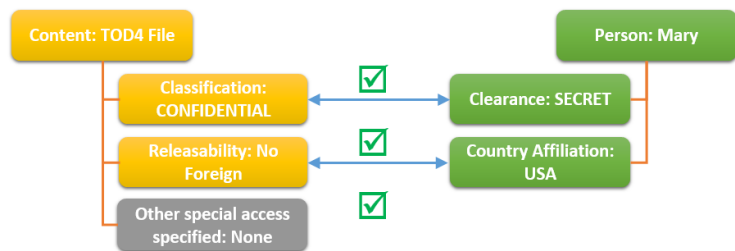


Figure 15: Security Authorization via Attribute Criteria (Granted Example)

For an access denied example, Bob is a liaison officer from the UK. He has a Top Secret Clearance. Bob will not be granted access to GET or LIST TOD4, as shown in Figure 16. If the TOD had been releasable to FVEY, Bob would have been authorized to access the data.

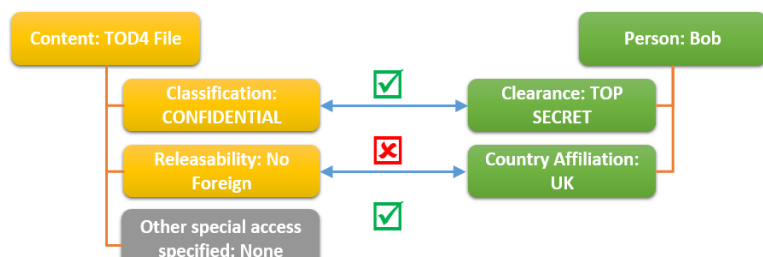


Figure 16: Security Authorization via Attribute Criteria (Denied Example)

4.7.2 Access by Role

Entitlements for restricted datasets are usually handled by specifying a “Group” that has access to the dataset, then confirming that users are group members. These groups are defined by roles. Roles usually are used for MANAGE access where there is a smaller set of users who can be provisioned into the group on an individual basis. They can also be used for READ access if there are special restrictions in place that cannot be covered by the authorization attributes. This role paradigm serves the entitlement group based on: Proper Use Memorandum (PUM) controls, restrictions based on intellectual property, international regulations or some other agreement binding on NGA.

The entitlements groups concept has four parts: roles, privileges, data (e.g., Content-IDs), and entities associated with the role. With these four elements, illustrated in Figure 17, the corresponding policy and rules set for any entitlements group based access can be automated.

Data Custodians and Stewards will use Entitlement Management Services to create the roles, and define the privileges, data and entities that are associated to the roles. Changes might take minutes, and would apply to any service or system using that data.

The following examples illustrate entitlements groups for two different use cases: 1) statutory restrictions, and 2) PUM.

In the statutory restrictions use case, because of international rules and regulations governing Aeronautical Safety of Navigation, only aeronautical analysts are allowed to access the Aeronautical production database and manage the data. Figure 18 shows how this group would be set up.

In use cases like Hurricane Katrina, where a PUM was needed to handle domestic imagery access, Figure 19 outlines how the data is handled by group entitlement role attributes.

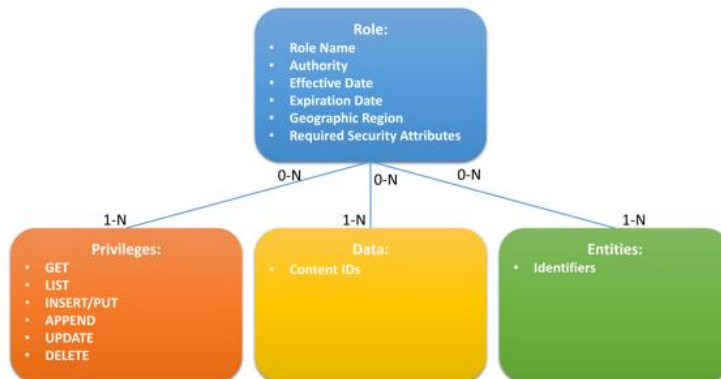


Figure 17: Entitlement Groups Concept

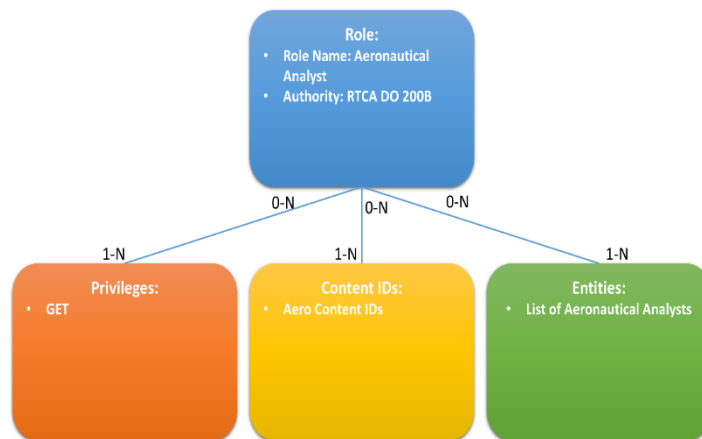


Figure 18: Statutory Restrictions (Use Case)

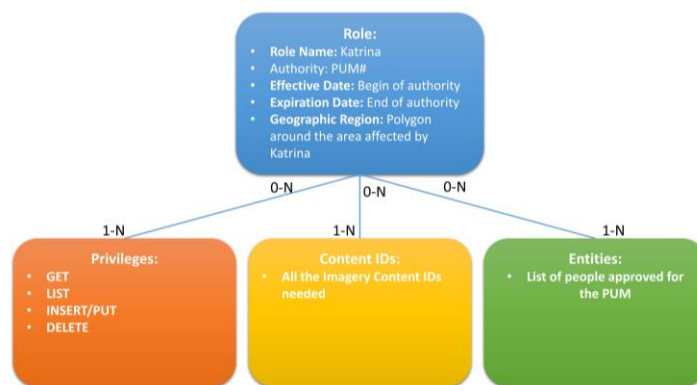


Figure 19: PUM Entitlement Group (Use Case)

No matter the reason for needing an entitlement group, a role can be defined, and privileges, data and entities assigned to it. This model can replace the custom authorization/role capabilities that were traditionally custom-created for each legacy system. With this model, NGA can create an enterprise Role Based Access Control (RBAC) capability that can be used by all systems.

4.8 Current State of Automated Data Access

GEOAxis provides the enterprise PDP capability for NGA automated data access. This PDP can perform the comparison of person attributes to the data attributes for authorization. GEOAxis provides a role mechanism that keeps the association of individuals and makes that available to applications.

The Data Inventory is where all Content-IDs for NGA data are defined, but the Content-IDs are not yet associated with the roles. The DMWG has surveyed the Data Custodians to get an initial list of data entitlements for READ and MANAGE data privileges against the Content-IDs.

Data is being migrated to the ODS, but the majority has not yet been conditioned and the architecture needed to achieve automated data access is not yet in place.

The path forward is based on file data and database data. The S3 Data Access Roadmap outlines the way to store file data in S3 and the Database Data Access Roadmap for database data.

4.8.1 S3 Data Access Roadmap

Vision. It is the intent to stand up an Enterprise Metadata Catalog that would have a record of each file stored in one of the S3 buckets in the Open Data Store. These records would include the security metadata both on the record itself and the data contained. The record security metadata would determine which results a person could see returned in a metadata catalog query (LIST) while the data security metadata would determine what data the user could download (GET). The S3 data access use case is demonstrated in Figure 20. The envisioned steps for gaining S3 bucket data access follow:

1. A user would submit their user credentials (PKI cert) to access the application.
2. The application would authenticate the user with the PDP. Along with the validation of the user, a Security Token Service (STS) token for that user is returned. That token is stored by the application for the session and is how the identity of the validated user is passed to the data services.

3. The user initiates a search request on the Metadata Catalog in the ODS. The web application passes the STS token and the search request to the Search API.
4. The Search API uses the PEP to pass the user token to the PDP to determine what types of data the user is allowed to access. The PDP returns the policy constraints to the PEP.

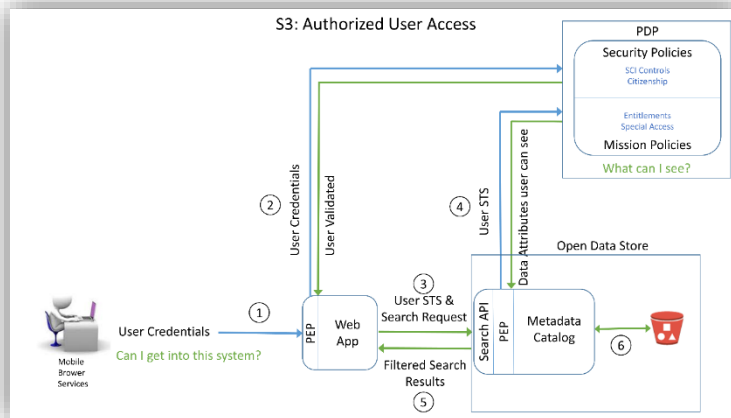


Figure 20: S3 Data Access Flow (Vision)

5. The PEP applies filters to the search request such that data the user cannot access is not even returned in the results. The filtered results are returned to the web application and to the user.
6. The user selects the data they want to access from the results list and executes a GET. This passes back through all the layers with the Metadata Catalog, PEP, and PDP to ensure the user is only getting what they are allowed to get (i.e., making it so an ineligible user who has acquired a link cannot get the data) and the files are disseminated to the user.

Current State. Trusted activity access will be phased out as the S3 data access capabilities mature. The current steps for gaining ODS data access, as shown in Figure 21, follows:

1. A user would submit their user credentials (PKI cert) to access the application.
2. The application would authenticate the user with the PDP.
3. The web application has full access to S3 buckets and Content-IDs it has been granted access to. There is no STS token being passed. There is no enterprise method for checking user attributes against data attributes.

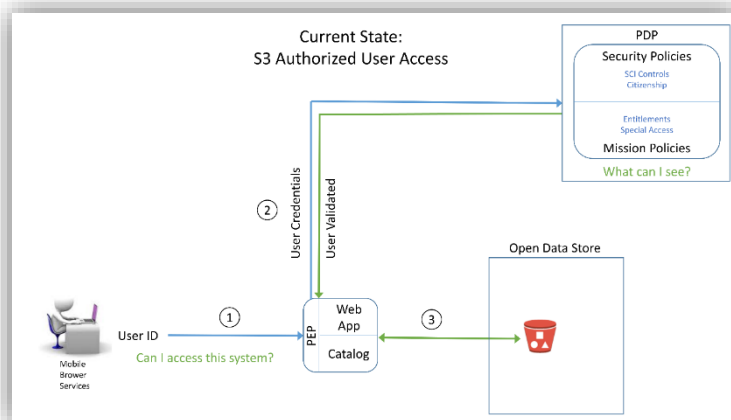


Figure 21: S3 Data Access for Trusted Activities

GEOAxis provides the enterprise PDP capability. The Metadata Catalog and associated PEP are in the process of being built, but not yet complete and exposed to the enterprise.

This means that activities cannot use the ODS Search API, PEP and Metadata Catalog. This necessitates allowing activities trusted access directly to the S3 buckets and virtual folders to READ and/or MANAGE the data.

It is then up to those activities with trusted access (i.e., trusted activities), to ensure that their users only have access to the appropriate data. In other words, they would operate much as they do now, with the exception that their data is stored in the ODS.

Until such time that the ODS Search API, PEP and Metadata Catalog are mature and exposed to the enterprise, the only access to data will be via the trusted activities, which in most cases will be the legacy systems.

Going Forward. The ODS Search API, PEP and Metadata Catalog will be matured, with capabilities released in phases as they are available. A Phase I solution to the Vision, for instance, might have a PEP/PDP combination that only has the ability to compare person attributes for Clearance, Country affiliation and SCI controls to grant basic GET and LIST entitlements. This would restrict the ODS services to only serving data out, via the Enterprise Services, that could be released based on those attributes.

A Phase II solution will enable a PEP/PDP combination that could handle role based entitlements. This would not only expand the data with READ privileges, but would enable the PDP and PEP to broker MANAGE privileges as well.

As the enterprise solutions in the ODS are able to perform the brokering on both READ and MANAGE privileges, activities will be required to use the enterprise services. Legacy activities that will persist in the long term will be required to shift over to use Enterprise Services.

4.8.2 Database Data Access Roadmap

Vision. The intent is to support fine-grained access control of data within databases. To do this, security metadata, as defined by the IC ISM standards, must be present in the database at the very least at a row/object/document level, and, ideally, at an attribute/key/value pair level.

The IC ISM standard has been incorporated into the NMF, and into the latest version of the NAS. So, databases whose schema is IC ISM compliant should be able to implement/use enterprise PEP/PDP services for fine-grained data access control on their database. The envisioned database data access is demonstrated in the steps outlined in Figure 22.

1. A user would submit their user credentials (PKI cert) to access the application.
2. The application would authenticate the user with the PDP. Along with the validation of the user, an STS token for that user is returned. That token is stored by the application for the session and is how the identity of the validated user is passed to the data services.
3. The user initiates a search request on the database in the ODS, using the web application. The web application passes the STS token and the search request to the Search API.
4. The PEP applies filters to the search request such that data the user cannot access is not even returned in the results. The filtered results are returned to the web application and to the user.

Note that the STS token is used differently in this example vice the S3 example. In the S3 example, the STS token was just used to pass the authenticated user identity, and a second query to the PDP was used to apply the policy constraints.

In this example, the first query to the PDP is not only getting the validation of the user, the data attributes the use case see – such that a second query is not needed to create the filter. The NRO, DNI and NGA are still working on their final answer to bulk authorization and which method will be used will be dependent on those decisions. Future guidance will contain any refinements/updates based on those decisions.

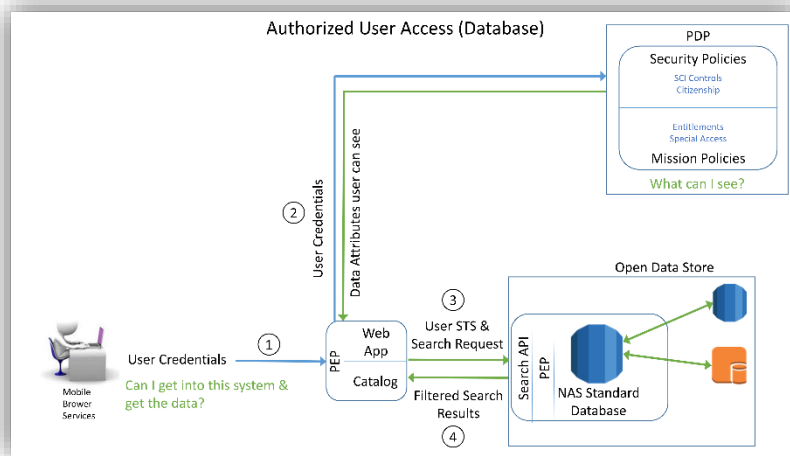


Figure 22: Database Access Flow (Envisioned)

Current State. There are a large number of legacy production and dissemination databases. While some of the legacy activities use GEOAxIS for authentication, almost none of the databases do any sort of authentication. Almost none of them use the PDP for authorization. Most have their own custom version of data access control. Most legacy applications and their databases are tightly coupled and any change in schema to accommodate the IC ISM would require considerable effort.

Currently, the only way to access the legacy databases, whether in the Data Center or in the ODS, is via the legacy application that manages that data.

Going Forward. Realizing that transforming all of the legacy databases and their associated legacy application would be “a bridge too far”, a NAS compliant database for NGA content is being set up that has previously been mapped to the NAS. This would work as a dissemination database, with the output of the various legacy databases being propagated to the enterprise NAS-based database. This could either be a physical database into which the production data is loaded, or a data virtualization capability that performs on the fly transformations to the NAS standard. This would enable both automated enterprise data access with fine-grained control, as well as enhanced discovery using the NSG standard definitions of entities and attribute found in the NAS.

For content that has not previously been mapped to the NAS, but would be appropriate to add to the NAS, the CDO office will work with the DC and the NSG Standards office to get the content incorporated.

For content that is not appropriate for the NAS, but would still like to take advantage of the automated, fine-grained access control, the CDO office will work with the DC to get the IC ISM incorporated into their schema.

4.9 Impacts

4.9.1 Application Development

With the stated vision for data access and handling, the obvious question is: “How it will impact future application development?” There will be two main effects on application development:

- Applications will not have to build their own authentication and authorization, and create their own logins and roles. Developers will be able to use enterprise authentication and authorization services.
- Applications will have to be able to propagate identity via STS tokens back to the data services to get the correct results. In the past, identity often ended at the front door of the applications, with all other actions taken by the application as an NPE.

4.9.2 Enterprise Utility

The impacts at the enterprise level are even greater than those at the development level:

- Removing the need for individual systems to create their own log-ins means that a single sign-on enterprise is possible.
- Removing the need for individual systems to implement their own entire authorization capability enables enterprise-based micro services.
- Identity propagation by all the applications and services involved in any transaction would make it much easier to perform audit logging.
- Data access based on attributes and roles, with data properly metadata tagged, removes the need to have separate networks and storage for groups/people with different access attributes (e.g., foreign partners).

APPENDIX A: DEFINITIONS

Term	Definition
Activity	"Activity" replaces the words application, system, entity, capability and program.
Activity Data	This is the technical data needed to make an activity/service run. This excludes mission content. This includes code, configuration files, logs, etc.
Analytic Registry and Repository/Catalog	The analytic registry will store artifacts software logic (such as XML Schema Definition documents, Web Service Description Language documents, and Service Level Agreements) related to analytic services and web enabled application interfaces (i.e., widgets and apps). Services and other software applications will use this repository at run time to search for specific analytic services and to retrieve service endpoints for invoking those services. Content producers will add and modify entries in the service registry as part of the process for service-enabling their content. An analytic catalog or repository containing human-readable descriptive information (metadata) about analytic services will serve as the user-facing repository that content producers, analysts, data scientists, and others use to discover information on analytic services. The analytic repository or catalog is conceptually distinct from the analytic registry but may be physically integrated with the registry implementation. The Analytic Registry can be part of the Service Registry and Repository/Catalog.
Analytical Data	This is the ephemeral working data/products that Analysts are analyzing on their desktops on in an Analytical Tool/Service. It is not considered Mission data until it is finalized/published to the gold repository and/or the customer. The published product created from the results of Analytical Data are an example of Mission data.
Archive	Stored infrequently used data,(e.g., AWS Glacier data objects or set of objects identified by an AWS Archive ID; archives are stored in an AWS Glacier Vault.)
Big Data	Refers to massive datasets and combinations of datasets with size and complexity beyond the ability of traditional software and tools to capture, store, manage, and analyze. As distinct from "Large Data", which refers to individual files, each of which is very large.
Conditioning (data)	Conditioning is the research and addition of metadata to the catalog to support enterprise discovery.
Content	Data, products, services, and knowledge that are discoverable and accessible. In AWS, any data in any format.
Content Collection Catalog	The Content Collection Catalog will provide a directory of the types of content available within the NGA enterprise. It will contain metadata describing each type of content, including its form, functions, access restrictions, suitability for use, inventory status, and associated points of contact. This catalog will serve as a tool enabling users to discover the kinds of content available to them. It will also be a means for enabling effective content governance, consistent web navigation of content, and compliance with content standards.
Corporate Data	Data used at the enterprise level that enables the business of NGA, but is not directly part of the mission. This would include PeopleSoft Data, GeoAxis Data, and other Enterprise Needs/Workflow/Process tools. This varies from the Conceptual Data model definition of corporate data in that activity data is excluded.
Dataset	A collection of data.
Data Lake	A large storage repository and processing engine that provides "massive storage for any kind of data, enormous processing power and the ability to handle virtually limitless concurrent tasks or jobs".
Managing Activity	The Activity (including systems or processes) that has primary management responsibility for a data set or content-ID. Note that there can be more than one Managing Activity for one content-ID.
Master Data Catalog	A subset of the Object Instance Catalog, it is the single source of basic business data used across all systems, applications and processes for the enterprise.
Metadata	Data about data; graphical or textual information about the content, quality, condition, origins, and characteristics of data.
Metadata Catalog	The Metadata Catalog is a human readable collection of metadata for file-based data.

Annex G: Cloud Data Guidance

Term	Definition
Mission data	Data used in performance of the NGA Mission, which would be considered “Gold Copy” or Record data. This would include source data (<i>host nation publications, imagery, commercially purchased datasets etc...</i>), the actual production data stores (ex: <i>Aeronautical Migration System (AMS) Databases Aeronautical Obstruction Environment (AOE) and Aeronautical Digital Data Environment (ADDE), Topographic Features Data Management (TFDM) Topographic Data Store (TDS)</i>) and data stores (ex: <i>Web Digital Vertical Obstruction File (WebDVOF) Database</i>) / <i>products (Digital Nautical Chart (DNC), Intel Reports, Geodesy Models, Navigation Planning (NAVPLAN) Charts, Controlled Imagery)/publications (Flight Information Publication (FLIP), American Practical Navigator, Fleet Guides etc...</i>) that NGA disseminates. This is “the content”.
Object	Binary Large Object (BLOB), file or other data set, identified by a key. Also, representation of physical or conceptual objects and their activities.
Object Catalog	Contains entries describing discrete instances of objects (for both Mission and Corporate data). The object catalog is a metadata catalog in which discrete instances of object data are described by a single record (row) in the catalog database. The Object Catalog is conceptually distinct from the Metadata Catalog (a component of NGA’s enterprise search capability architecture) but may be physically integrated with the implementation of the Enterprise Metadata Catalog. Each object catalog entry will contain a Universally Unique Identifier (UUID) for an object instance, descriptive metadata to enable discovery and search, IC mandated metadata relating to security classification and access control, and a machine resolvable link (such as URL (Uniform Resource Locator)) to enable access to the object.
Organizational Data	A subset of Corporate data – files and data organizations generally put in their shared organization drives now (a migration-specific category). This is data needed for the business process of individual organizations. This would include things like International Standards Organization (ISO) processes, Org Charts, briefings... the types of things organizations generally put in their shared organization drives or SharePoint.
Product	Data that is a packaged result of NGA effort, as opposed to raw data from outside sources.
Scalability	Data transaction or storage amount increases or decreases with performance neutrality.
Semantics Catalog (Controlled Vocabulary (CV) Registry)	A directory of pre-defined vocabulary schemas including taxonomies, synonym lists (thesauri), approved keyword lists, and other knowledge organization schemas (ontologies). The controlled vocabularies will be searchable and accessible for use in content preparation activities (such as metadata tagging), providing user selections in search query construction, and enabling semantic search capabilities. One of the principal taxonomies in the current NGA enterprise is the NSG content taxonomy that defines hierarchical categories describing the primary form and mission of each type of NGA managed content. The NSG content taxonomy will be the basis for identifying and describing entries in the Content Collection registry.
Service Registry and Repository/Catalog	The service registry will store artifacts (such as XML Schema Definition documents, Web Service Description Language documents, and Service Level Agreements) related to enterprise web services and web-enabled application interfaces (i.e., widgets and apps). Services and other software applications will use this repository at run time to search for specific web services and to retrieve service endpoints for invoking those services. Content producers will add and modify entries in the service registry as part of the process for service-enabling their content. A service catalog or repository containing human-readable descriptive information (metadata) about web services will serve as the user-facing repository that content producers, system developers, and other users can use to discover information on enterprise web services. The service repository or catalog is conceptually distinct from the service registry but may be physically integrated with the service registry implementation.

APPENDIX B: REFERENCES

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18. "Pentaho, Hadoop and Data Lakes", James Dixon's Blog
19. Intelligence Community Technical Specification, Abstract Data Definition for Electronic Records Management, Version 2014-DEC
20. NGA Standardization Document National System for Geospatial Intelligence Metadata Foundation (NMF) – Part 1: Core (2014-09-23) Version 2.2
21. NGA Standardization Document National System for Geospatial Intelligence Metadata Foundation (NMF) – Part 4: Records Management Metadata (2015-04-03) Version 2.0
22. NGA Data Management Guidance, 2016
23. NGA Unified Data Strategy (UDS), 2015.
24. NGA OpenDataStore Concept of Operations, Version 0.3, May 2017.

APPENDIX C: NMF MANDATORY FIELDS

C.1 NGA NMFv3 Mandatory Fields

The following fields are the minimum required fields specified by [NMF Version 3.0](#). They are listed in the order they were specified. These fields are designed to support the several data categories required for file based discovery in the cloud.

NMF metadata that is included in a TDF package is added to the portions of the XML documentation that are referred to as the “other or mission assertion” related elements.

NMF v3.0 Required Fields		
Metadata Tag	Description	Domain Guidance
Metadata Point of Contact	The party or parties acting in a role of responsibility for a set of resource metadata.	The first instance of this element is to have the role of “creator”. Resource Metadata (Table 7) Metadata Contact – Populate using the Responsibility datatype (Table 46) with role (Table 3) of ‘PoC’ and identifying the individual or organization responsible for the metadata record.
Metadata Date	The date(s), and optionally time(s), of an event involving a set of resource metadata.	The first instance of this element is to have the role of “creation”. Resource Metadata (Table 7) Resource Metadata Date – Populate using Date datatype (Table 50) with the date and DateType (Table 3) such as ‘creation’, ‘publication’, or ‘revision’ for when the metadata record was created, published, or revised.
Metadata Standard Title	The name of the metadata standard which determines the structure and content of this set of resource metadata.	Resource Metadata (Table 7) Metadata Standard Citation – Populate using Citation datatype (Table 45) identifying the Cited Resource Title of the standard used for the metadata record. NSG Metadata Foundation (NMF), Version 3.0 10
Metadata Standard Edition	The version of the metadata standard which determines the structure and content of this set of resource metadata.	Resource Metadata (Table 7) Metadata Standard Citation – Populate using Identifier datatype (Table 54) identifying the code for the edition of the standard used for the metadata record.
Metadata Scope Code	A brief indicator of the type of resource for which metadata information is reported.	Metadata Scope (Table 8) Resource Scope – Populate using ScopeCode codelist (Table 3) to identify the type of resource the metadata applies to such as ‘dataset’, ‘series’, ‘attribute’, or ‘feature’.
Metadata Scope Name	A word or phrase that describes the type of resource for which information is reported.	Metadata Scope (Table 8) Resource Scope – Populate using Resource Scope Name identifying type of resource if Resource Scope is not ‘dataset’.
Metadata Classification	The classification level of the metadata, in accordance with the Intelligence Community (IC) Security Markings Manual.	Security Attributes Group (Table 33) Resource Classification – Populate with ResClassificationStrucText datatype (Table 33).

Metadata Classification System	The classification system of the metadata, in accordance with the Intelligence Community (IC) Security Markings Manual.	Security Attributes Group (Table 33) Resource Owner-Producer – Populate with ResOwnerProducerStrucText datatype (Table 33).
Resource Title	The name by which a cited resource is known.	Identification (Table 9) Resource Citation – Populate with using Cited Resource Title (Table 45).
Resource Abstract	A brief statement or narrative summary of the resource.	Identification (Table 9) Resource Abstract – Populate with free text providing a short description of the resource.
Resource Point of Contact	The party(ies) acting in a role of responsibility for the resource.	The first instance of this element is to have the role of “originator”. Identification (Table 9) Resource Point of Contact – Populate using the Responsibility datatype (Table 46) with role (Table 3) of ‘PoC’ and identifying the individual or organization responsible for the metadata record. NSG Metadata Foundation (NMF), Version 3.0 11
Resource Date	A reference date for a cited resource. The first instance of this element is to have the role of “creation”.	Identification (Table 9) Resource Citation – Populate using Date datatype (Table 50) with the date and DateType (Table 3) such as ‘creation’, ‘publication’, or ‘revision’ for when the metadata record was created, published, or revised.
Resource Identifier	A value uniquely identifying the resource within a namespace.	Identification (Table 9) Resource Citation – Populate using Identifier datatype (Table 54) identifying the code for the unique resource identifier.
Resource Geographic Location	The spatial extent of the resource. The spatial extent is a geographic identifier (for example: a country name), a bounding box (for example: the bounding latitudes and longitudes), or a bounding object (for example: a set of coordinate points).	Identification (Table 9) Resource Extent – Populate using Geographic Extent datatype (Table 39) to specify a Geographic Bounding Box, Geographic Description, or Geographic Bounding Object.
Resource Language	Designation of the locale language.	Data Identification (Table 10) Text Locale – Populate using Language Locale (Table 55) LanguageCode codelist (Table 3) to identify language of resource.
Resource Character Set	Designation of the character set to be used to encode the textual value of the locale.	Data Identification (Table 10) Text Locale – Populate using Locale Character Encoding (Table 55) IANA Charset codelist (Table 3) to identify character set used in resource.
Resource Topic Category Code	A theme or topic keyword that represents a subject of the resource.	Identification (Table 9) Topic Category – Populate using MD_TopicCategoryCode enumeration (Table 4).
Resource Keywords	Information about keywords describing this resource.	Keywords (Table 12) Keywords – Populate with Keyword to identify content of the resource.
Resource Classification	The classification level of the resource, in accordance with the Intelligence Community (IC) Security Markings Manual.	Security Attributes Group (Table 33) Resource Classification - Populate with ResClassificationStrucText datatype (Table 33).
Resource Classification System	The classification system of the resource, in accordance with the Intelligence Community (IC) Security Markings Manual.	Security Attributes Group (Table 33) Resource Owner-Producer – Populate with ResOwnerProducerStrucText datatype (Table 33). NSG Metadata Foundation (NMF), Version 3.0 12

Resource Category	The particular category of a data resource within a defined taxonomy.	Resources within a category have similar information content and have been produced by similar processing methods. The default is a resource category type of "other". NMF Data Identification (Table 11) Data Identification – Populate with Resource Category using code from ResourceCategoryCode codeList (Table 3).
Metadata Language	Designation of the locale language.	Obligation: Required when not defined by encoding. Data Identification (Table 10) Text Locale – Populate using Language Locale (Table 55) LanguageCode codeList (Table 3) to identify language of resource.
Metadata Character Set	Designation of the character set to be used to encode the textual value of the locale.	Data Identification (Table 10) Text Locale – Populate using Locale Character Encoding (Table 55) IANA Charset codeList (Table 3) to identify character set used in resource.
Parent Metadata Citation	A standardized reference to a set of metadata that is in a parent relationship to this set of resource metadata.	Obligation: Required if there is an upper scope Level. Identification (Table 9) Parent Metadata Citation – Populate using Citation datatype (Table 45) identifying the identifier of parent metadata.
Resource Temporal Extent	The time period covered by the resource	Obligation: Required when Resource Extent Description, or Geographic Extent, or Vertical Extent not documented. Temporal Extent (Table 43) Temporal Extent – Populate using Temporal Geometric Primitive (Table 60), such as Temporal Period or Temporal Instance.
Resource Coordinate Reference System	Information about a spatial or temporal reference system used by representations in the resource.	Obligation: Required if the resource includes coordinates. Reference System Information (Table 37) Reference System Identifier – Populate with Identifier (Table 54) Code identifying the reference system used in the resource.

C.2 NGA Analysis and Source Mandatory Fields

The NGA Source organization provided the following fields that are deemed essential and necessary for file based discovery and these fields are provided as a requirement for essential metadata file based discovery in the cloud.

<i>NGA Analysis and Source Mandatory Fields</i>	
<i>Metadata Tag</i>	<i>Definition</i>
Abstract (Resource)	Brief narrative summary of the content of the resource(s)
Classification	Classification of material to be published
Classification System	Name of the classification system
Content (Resource) POC	Party (Person or Organization) responsible for the resource
Date (of Cited Source)	Date for the cited resource
Date Posted (or Published)	Date that the resource data was posted or Published
Dissemination Controls	One or more indicators identifying the expansion or limitation on the distribution of information
Geographic Location (Extent)	Location that the published material covers
Identifier (Resource)	Alphanumeric value identifying the resource
Keyword	Specified by a controlled-vocabulary code or by keywords.
Metadata Date Stamp	Date that the metadata was created
Metadata Language	Language used for documenting metadata
Metadata POC	Party (Person or Organization) responsible for the metadata information
Metadata Standard Name	Name of the metadata standard (including profile name) used
Metadata Standard Versions	Version (profile) of the metadata standards used
Releaseability	Information explicitly marked for appropriate foreign disclosure or release
Resource Language	Language(s) used within the dataset
Review Date (Not used by A)	Date the material is to be reviewed for accuracy (one year after published date)
Resource Character Set	Full name of the character coding standard used for the dataset
Resource Originator POC	Party (Person or Organization) who created the resource
Resource Topic Category	Main theme(s) of the dataset
Temporal Extent	Information about the temporal extent of the resource
Title (Resource)	Name by which the cited resource is known
Online Resource	Information about on-line sources from which the resource can be obtained

C.3 NGA X Organization Mandatory Fields

The NGA X organization provided the following list of fields that are deemed essential and necessary for file based discovery. These fields required and essential metadata for file based discovery in the cloud.

<i>NGA X Organization Mandatory Metadata</i>	
<i>Metadata Tag</i>	<i>Definition</i>
Nominator Name	Indicates user who nominated the object
Nominator Email	Indicates email of nominator
Content URL	Provides reference location to object
Content Name	Indicates object name
Keywords	Metadata that describes object in one word
Content Description	Informs potential user of data object's subject
Content Mission	Indicates object's mission association
Licensing Agreements/Restrictions	Indicates legal usage of object
Which security network I find this product on	Provides reference location to object by security network
Maintenance Cycle/Update Frequency	Indicates how often object is updated
Data Type	Indicates data type (file type)
Event or NGA Crisis Action Team	Indicates whether or not the data supports an Event or NGA Crisis Team
Content Owner Name	Indicates who can be questioned about the object
Organization/Office/Branch	Indicates the Org./Office/Branch that the Content Owner supports
Organization/Office/Branch Email	Indicates Organization/Office/Branch Email – Contact Information
Organization/Office/Branch Phone	Indicates Organization/Office/Branch Phone – Contact Information

APPENDIX D: NMF CROSS REFERENCE WITH EDH AND TDF

<i>NMF Cross Reference with EDH and TDF</i>		
<i>NMF Metadata Tag</i>	<i>EDH Mapping</i>	<i>TDF</i>
Resource Date	Create Date	Applies to Payload
Metadata Point of Contact	Responsible Entity	Applies to Mission Assertion
Metadata Date Stamp	Create Date	Applies to Mission Assertion
Metadata Security Classification	ARH/ISM/Classification	Applies to Handling Assertion
Metadata Security Classification System	ARH/ISM/Owner Producer	Applies to Handling Assertion
Resource Security Classification	ARH/ISM/Classification	Applies to Payload
Resource Security Classification System	ARH/ISM/Owner Producer	Applies to Payload
Resource Identifier	Identifier	Applies to Payload
Resource Originator	Responsible Entity	Applies to Payload
Resource Point Of Contact	Responsible Entity	Applies to Payload

APPENDIX E: IC MANDATORY FIELDS

E.1 Electronic Records Management (ERM)

Electronic Records Management (ERM) is an IC mandatory standard and is required to provide the means to audit, track, and manage disposition information. The ERM specification is found within the IC CIO's [Data/Technical Specifications](#) library.

ERM has dependencies with the TDF, ISM and IC-EDH standards. Dependencies and the ERM metadata tags are shown in the following Table:

<i>ERM Dependent Standards</i>			
<i>Abbreviation</i>	<i>Category</i>	<i>ID</i>	<i>Title</i>
TDF	Packaging/Transport	IC-TDF.XML.V2014-DEC	XML Data Encoding Specification for Enterprise Data Header, Version 2015-AUG
ISM	Information Security	ISM.XML.V2015-Aug	XML Data Encoding Specification for Information Security Markings, Version, 2015-AUG
IC-EDH	Information Security	IC-EDH.XML.V4	XML Data Encoding Specification for Enterprise Data Header, Version 2015-AUG

<i>ERM Standard</i>	
<i>Metadata Tag</i>	<i>Description</i>
erm:AppliedBy	Individual that performed the disposition of the Record.
erm:Authorizer	Entity responsible for the hold.
erm:DateApplied	Date when action was taken or applied to the Record.
erm:DateEligible	Date the record is eligible for disposition.
erm:DateLimit	Date by which a disposition action must be taken regarding the Record.
erm:Dispositoin	An action taken with regard to U.S. federal records that are no longer needed for current government business as determined by their appraisal pursuant to legislation, regulation, or administrative procedure.
erm:EffectiveDate	Date the hold order was effective.
erm:ElectronicRecordsManagementMetadata	Information used by a records management system to manage a resource. A records management system systematically controls the creation, distribution, use, maintenance, and disposition of recorded information maintained as evidence of business activities and transactions.
erm:FoiaOpsIndicator	Indicator stating if the Record is exempted from FOIA search and publication (e.g. operations exemption at CIA): each agency has specific exemptions they request and maintain.
erm:Hold	A suspension of disposition of a set of one or more records in the records management process of the cognizant organization or institution.
erm:Identifier	Specific way to identify the hold (unique among hold types and agencies).
erm:Justification	Legal, policy, or mission driver for the hold order.

erm:OfficeOfRecord	Identifies the Agency and organization element within the agency that is responsible for making decisions related to the Record. Also identifies which organizational element within an agency owns/is responsible for the official copy of the Record.
erm:RecordControl	Identifier for Agency Record Control System (RCS) disposition policies (per NARA).
erm:RecordDesignationDate	Date the Record is declared final (e.g. cutoff, publication, creation) and starts the retention period which will be used to calculate the disposition date based on the retention schedule.
erm:ReleasedDate	Date the hold order was released.
erm:ReviewIndicator	Indicator stating that the resource has been reviewed to determine the appropriate RCS, if applicable.
erm:Type	The category of a hold on a managed Record's disposition.
erm:VitalRecordsIndicator	Indicator stating that the Record is a Vital Record. Vital Records are those Records considered essential to the continuity of operation during and after emergencies or disaster conditions. Also known as Essential Records (per Federal Continuity Directive 1 (FCD 1))

E.2 Trusted Data Format (TDF)

Trusted Data Format (TDF) is an IC mandatory standard and is identified as the way ahead strategy for implementing cloud-based information exchange and discovery on the IC enterprise. It is the metadata envelope or wrapper to transmit a data object or a reference to a data object.

The TDF specification is found in the following location within the [Data/Technical Specifications](#) library. TDF has dependencies with the standards show in the following Table:

<i>TDF Dependent Standards</i>			
<i>Abbreviation</i>	<i>Category</i>	<i>ID</i>	<i>Title</i>
ISM	Information Security	ISM.XML.V2015-Aug	XML Data Encoding Specification for Information Security Markings, Version, 2015-AUG
NTK	Information Security	NTK.XML.V2015-AUG	XML Data Encoding Specification for Need-to-Know Metadata, Version 2015-AUG
IC-EDH	Information Resource	IC-EDH.XML.V2015-AUG	XML Data Encoding Specifications for Enterprise Data header, Version 2015-AUG
ARH	Information Security	ARH.XML.V3	XML Data Encoding Specification for Access Rights and Handling, Version 3, 6 September 2013
RevRecall	Information Resource	REVRECALL.XML.V1	XML Data Encoding Specification for Revision Recall, Version 1, 10 March 2014

The following Table provides the description and business rules definitions of the primary TDF concepts (i.e., TDO, TDC, assertions, and payload).

Concept	Description and Business Rules
Trusted Data Object (TDO)	A Trusted Data Object (TDO) is the basic building block and contains one or more assertions (statement about the data) and the payload data. A TDO can be used at the very minimum to specify payload and assertions for a single payload item. The basic structure of a TDO is the following: <pre> <TrustedDataObject> <tdf:HandlingAssertion tdf:scope="TDO"> <tdf:HandlingStatement> <edh:ExternalEDH/> <arh:Security..> <ntk:Access.../> <arh:Security../> </edh:ExternalEDH/> </tdf:HandlingStatement/> </tdf:HandlingAssertion> <tdf:Assertion tdf:scope="PAYL"> <tdf:ReferenceValuePayload tdf:uri="http://example/reference/payload"/> 1 (one) </tdf:Assertion> </TrustedDataObject> </pre> 1. Note the use of the HandlingAssertion scope attribute to refer the scope of the handling assertion for the "TDO". 2. Note the use of the Assertion scope attribute to refer to the scope of the assertion for the "PAYL". 3. Note the use of the ExternalEDH element to refer to an external ... 4. Note the use of ReferenceValuePayload element to refer to an external reference to the payload item.
Trusted Data Collection (TDC)	A Trusted Data Collection (TDC) is a collection of Trusted Data Objects (TDOs) used to assemble more than one payload item that can be related or unrelated and contains one or more The basic structure of a TDC is the following: <pre> <TrustedDataCollection> <tdf:HandlingAssertion tdf:scope="TDC"> <tdf:HandlingStatement> <edh:ExternalEDH/> <arh:Security..> <ntk:Access.../> <arh:Security../> </edh:ExternalEDH/> </tdf:HandlingStatement/> </tdf:HandlingAssertion> <TrustedDataObject> <tdf:Assertion/> </TrustedDataObject> </TrustedDataCollection> </pre> 1. Note the use of the HandlingAssertion scope attribute to refer the scope of the handling assertion for the "TDC".

Concept	Description and Business Rules
Handling Assertion	A handling assertion is a statement about the access, rights, and handling instructions of the payload. <i>As shown in the TDO structure example above,</i> - At least one handling instruction is required for each TrustedDataCollection (TCO) or TrustedDataObject (TDO) - An EDH and related security attributes are included in a handling assertion. - Definitions of EDH and related security attributes are defined in the EDH section. A handling assertion: - is never encrypted. - can include a Revision Recall handling assertion. - is made before other assertions in the document order <i>as shown in the TDO and TCO examples</i>. - can include XML, binary, and by reference, formats.
Handling Assertion Scopes	A handling assertion is identified by the tdf:HandlingAssertion element and will contain an attribute scope (@type:scope). The scope attribute of the handling assertion will specify the scope of the handling assertion ("TDC", "DES_TDO", "DESC_PAYL"). Additional information for assertion scopes is found in Section 2.3 of the specification. See section 2.3.1.2 Assertion Scopes within the TDC for additional scopes and definitions.
Other Assertion or Mission Assertion	Other assertions or mission assertions are a statement about the payload that are not handling assertions and relate to the mission, discovery, or task order, etc... The Assertion attribute type is used to indicate the category of assertion that is intended, such as <Assertion @type="Discovery"/>
Assertion Scopes (Mission and Other Assertions)	An assertion is identified by the tdf:Assertion element and will contain an attribute scope (@type:scope). The scope attribute of the handling assertion will specify the scope of the handling assertion ("TDO", "PAYL"). Additional information for assertion scopes is found in Section 2.3 of the specification. See section 2.3.1.2 Assertion Scopes within the TDC for additional scopes and definitions.
Payload	Payload information includes the following formats: string, XML, binary, and by reference. The following TDF elements are used to convey the different payload: - StringPayload - Base64BinaryPayload - ReferenceValuePayload - StructuredPayload

A sequential output of the primary TDF XML elements is shown in the following Table:

TDF XML	
TDF Element	Description
{tdf} IC-TDF-collection	The root element of a Trusted Data Collection. A TDC can have a single handling assertion containing and IC-EDH specification and its scope must be TDC. There may also be an optional second handling assertion scope [TDC] that contains Revision/Recall information for the TDC using the RevRecall.XML specification. A handling assertion must not be encrypted.
{TrustedDataCollection} tdf:HandlingAssertion	A specific type of assertion designed to be used for access, rights, and handling instructions. It is expected that handling instructions should never have metadata about themselves and they should never be encrypted. Therefore, unlike regular assertions, handling assertions do not support statement metadata or encryption.
{TrustedDataCollection, tdf:HandlingAssertion} attr:scope	The grouping of objects to which the assertion applies.
{TrustedDataCollection, tdf:HandlingAssertion} tdf:HandlingStatement	Intended for access, rights, and/or handling instructions that apply to the scope of the assertion.
{tdf} TrustedDataObject	A TDO has at a minimum two Handling Assertions: A TDO handling assertion and a payload handling assertion. This allows for separate access control decisions to be made for the payload versus the entire TDO (which includes payload metadata).
{TrustedDataObject} tdf:HandlingAssertion	A specific type of assertion designed to be used for access, rights, and handling instructions. It is expected that handling instructions should never have metadata about themselves and they should never be encrypted. Therefore, unlike regular assertions, handling assertions do not support statement metadata or encryption.
{TrustedDataObject} tdf: Assertion	Used to express metadata about the objects expressed in the scope attribute of the assertion. An assertion also supports metadata about the assertion statement for the purposes of indicating any handling instructions pertinent to the statement itself. Also supports encrypted statements and binding the statement with objects in its scope.
{TrustedDataObject, tdf: Assertion} attr:scope	The grouping of objects to which the assertion applies.
{TrustedDataObject, tdf: Assertion} tdf:StringStatement	Intended for textual statement content encoded as a string. Perhaps the contents of a text file.
{TrustedDataObject, tdf:Assertion} tdf:Base64BinaryStatement	Intended for holding base64binary statement values such as a file or other binary encoded data.
{TrustedDataObject, tdf:Assertion} tdf:Reference-ValuePayload	Used to reference payloads that are not embedded in the TDO but stored in a remote/external location.

E.3 IC Enterprise Data Header (IC-EDH)

IC Enterprise Data Header (EDH) is an XML encoding standard intended to allow interoperability and data sharing by establishing a core set of universal identity and categorization metadata attributes, for all stored data. As part of and along with the data exchange standard Trusted Data Format (TDF), it provides the common XML elements and structure needed in the enterprise to exchange and store information, including exchanges between systems and cloud based environments.

The IC-EDH specification is found in the following location within the [Data/Technical Specifications](#) library. IC-EDH has dependencies with the standards show in the following Table.

<i>IC-EDH Dependent Standards</i>			
<i>Abbreviation</i>	<i>Category</i>	<i>ID</i>	<i>Title</i>
IC-ID	Information Resource	IC-ID.XML.V1	XML Data Encoding Specification for Intelligence Community Identifier Metadata, Version 1, 10 April 2013
ARH	Information Security	ARH.XML.V3	XML Data Encoding Specification for Access Rights and Handling, Version 3, 6 September 2013
ISM	Information Security	ISM.XML.V2015-Aug	XML Data Encoding Specification for Information Security Markings, Version, 2015-AUG
NTK	Information Security	NTK.XML.V2015-AUG	XML Data Encoding Specification for Need-to-Know Metadata, Version 2015-AUG
ISMCAT	Information Security	ISMCAT.CES.V2016-May	XML Controlled Vocabulary Encoding Specification for ISM Country Codes and Tetragraphs, Version 2016-MAY
US Agency	Information Resource	USGOVAgency.CES.V2014-Sep	XML Controlled Vocabulary (CVE) Encoding Specification Government Agency Acronyms, Version 2014-SEP, 01 September 2014

The complete list of EDH elements and definitions are shown in the following Table. Note that the namespace for each of the representative standards is shown for the element along with the parent-child relationship.

<i>IC-EDH</i>	
<i>EDH Element</i>	<i>Description</i>
{edh:ExternalEDH} edh:ExternalEDH	Used for specifying an object's basic header information required for exchange on the IC Enterprise when the object is not in the instance document that this element is used. Values specified in this header will not contribute to roll-up.
{edh:ExternalEDH} ntk:DESVersion	e.g. 201508
{edh:ExternalEDH} ism:DESVersion	e.g. 201508
{edh:ExternalEDH} ism:ISMACTESVersion	e.g. 20105
{edh:ExternalEDH} arh:DESVersion	e.g. 3
{edh:ExternalEDH} edh:DESVersion	e.g. 201508
{edh:ExternalEDH} edh:DataItemCreateDate	e.g. 2006-05-04T18:13:51.0Z

IC-EDH	
EDH Element	Description
{edh:ExternalEDH} edh:AuthorizationReference	Attribute to hold the authorization reference of the data object referred to by the EDH. One or more indicators of a unique and documented legal basis for all activities surrounding the creation, retention and use of an information resource.
{edh:ExternalEDH} edh:AuthorizationReferenceType	One or more indicators of a unique and documented legal basis for all activities surrounding the creation, retention and use of an information resource.
{edh:ExternalEDH} edh:ResponsibleEntity	The creator entity that created the data item referenced by the EDH.
{edh:ExternalEDH, edh:ResponsibleEntity,ism:SecurityAttributesGroup} ism:atomicEnergyMarkings	Applicable atomic energy information markings for a document or portion This attribute is used at both resource and portion levels. The permissible values for this simple type are defined in the ISM Atomic Energy Markings CVE: CVEnumISMAtomicEnergyMarkings.xml. Applicable atomic energy information markings for a document or portion This attribute is used at both resource and portion levels. The permissible values for this simple type are defined in the ISM Atomic Energy Markings CVE: CVEnumISMAtomicEnergyMarkings.xml.
{edh:ExternalEDH, edh:ResponsibleEntity,ism:SecurityAttributesGroup} ism:classification	The highest or level of classification applicable to the containing document portion The Classification element is always used in conjunction with the Owner Producer element. Taken together, the two elements specify the classification category (TS, S, C, R, or U) and the type of classification (US, non-US, or Joint). This attribute is used to render portion marks and security banners. This attribute is used at both resource and portion levels. The permissible values for this simple type are defined in the ISM Classification All CVE: CVEnumISMClassificationAll.xml
{edh:ExternalEDH, edh:ResponsibleEntity,ism:SecurityAttributesGroup} ism:classificationReason	One or more reason indicators or explanatory text describing the basis for an original classification decision This attribute corresponds to the "Reason" line of a document's classification authority block, and it is only used, and only allowed, when classification is the result of an original classification decision. It is used primarily at the resource level.
{edh:ExternalEDH,edh:ResponsibleEntity,ism:SecurityAttributesOptionGroup} ism:classifiedBy	The identity, by name or personal identifier and position title, of the original classification authority for a document This attribute corresponds to the "Classified By" line of a resource's classification authority block. It is used primarily at the resource level.
{edh:ExternalEDH,edh:ResponsibleEntity,ism:SecurityAttributesOptionGroup} ism:compilationReason	The reason that the classification of the document is more restrictive than the simple roll-up of the marked portions of the document This attribute is an indicator that there is not accidental over-classification of the document. Users must exercise special care beyond that indicated by the portion marks when using this information.

IC-EDH	
EDH Element	Description
{edh:ExternalEDH,edh:ResponsibleEntity,ism:SecurityAttributesOptionGroup} ism:declassDate	The specific date when the resource is subject to automatic declassification procedures if not properly exempted from automatic declassification This attribute corresponds to the “Declassify On” line of a resource’s classification authority block. It is used primarily at the resource level.
{edh:ExternalEDH,edh:ResponsibleEntity,ism:SecurityAttributesOptionGroup} ism:declassEvent	A description of an event upon which the information shall be subject to automatic declassification procedures if not properly exempted from automatic declassification A description of an event upon which the information shall be automatically declassified if not properly exempted from automatic declassification. This attribute corresponds to the “Declassify On” line of a resource’s classification authority block. It is used primarily at the resource level.
{edh:ExternalEDH,edh:ResponsibleEntity,ism:SecurityAttributesOptionGroup} ism:declassException	The exemption from automatic declassification that is claimed for a document This element is used in conjunction with the Declassification Date or Declassification Event, and it corresponds to the “Declassify On” line of a resource’s classification authority block. It is used primarily at the resource level. The permissible values for this attribute are defined in the ISM N25X CVE: CVEEnumISMN25X.xml. ISOO Guidance: @declassException should be a SINGLE value giving the longest protection.
{edh:ExternalEDH, edh:ResponsibleEntity, ism:SecurityAttributesGroup} ism:derivativelyClassifiedBy	The identity, by name or personal identifier, of the derivative classification authority This attribute corresponds to the “Classified By” line of a resource’s classification authority block, and it is used primarily at the resource level.
{edh:ExternalEDH, edh:ResponsibleEntity,ism:SecurityAttriburesGroup} ism:derivedFrom	A citation of the authoritative source or sources of the classification markings used in a derivative classification decision for a classified document This attribute corresponds to the “Derived From” line of a document’s classification authority block, and it is used primarily at the resource level. ISOO Guidance: The source of derivative classification. (1) The derivative classifier shall concisely identify the source document or the classification guide on the “Derived From” line, including the agency and, where available, the office of origin, and the date of the source or guide. An example might appear as: Derived From: Memo, “Funding Problems,” October 20, 2008, Office of Administration, Department of Good Works or Derived From: CG No. 1, Department of Good Works, dated October 20, 2008 (i) When a document is classified derivatively on the basis of more than one source document or classification guide, the “Derived From” line shall appear as: Derived From: Multiple Sources (ii) The derivative classifier shall include a listing of the source materials on, or attached to, each derivatively classified document.

IC-EDH	
EDH Element	Description
{edh:ExternalEDH,edh:ResponsibleEntity,ism:SecurityAttributesOptionGroup} ism:displayOnlyTo	The set of countries and/or international organizations associated with a “Display Only To” marking The “Display Only To” marking indicates that a document is authorized for foreign viewing by appropriate affiliates of approved countries and/or international organizations without providing the foreign recipient with a copy for retention in any medium (physical or electronic). This attribute is used at both the resource and the portion levels. The permissible values for this attribute are defined in the ISM Rel To CVE: CVEnumISMRelTo.xml.
{edh:ExternalEDH,edh:ResponsibleEntity,ism:SecurityAttributesOptionGroup} ism:disseminationControls	Applicable dissemination control markings for a document or portion This attribute is rendered in portion marks and banners. The permissible values for this attribute are defined in the ISM Dissemination CVE: CVEnumISMDissem.xml
{edh:ExternalEDH,edh:ResponsibleEntity,ism:SecurityAttributesOptionGroup} ism:FGISourceOpen	The set of countries and/or international organizations whose information is derivatively sourced in a document when the source of the information is not concealed (also used for cases when the source is unknown) FGI markings protect foreign-owned or foreign-produced information and are applied based on sharing agreements or arrangements with the source country or organization. This attribute is used at both the resource and the portion levels. The permissible values for this attribute are defined in the ISM FGI Open CVE: CVEnumISMFGIOpen.xml. Note: the value “UNKNOWN” is allowed in cases where the source of the information is not known.

IC-EDH	
EDH Element	Description
{edh:ExternalEDH, edh:ResponsibleEntity, ism:SecurityAttriburesOptionGroup} ism:FGISourceProtected	The set of countries and/or international organizations whose information is derivatively sourced in a document when the source of the information must be concealed This attribute has specific rules concerning its usage: PROTECTED SPACES — Within protected internal organizational spaces, this attribute may be used to maintain a formal record of the foreign country or countries and/or registered international organization(s) that are the non-disclosable owner(s) and/or producer(s) of information which is categorized as foreign government information according to Security Markings Program guidelines. If the data element is employed in this manner, then additional measures must be taken prior to dissemination of the resource to shared spaces so that any indications of the non-disclosable owner(s) and/or producer(s) of information within the resource are eliminated. In all cases, the corresponding portion marking or banner marking should be compliant with Security Markings Program guidelines for FGI when the source must be concealed. In other words, even if the data element is being employed within protected internal organizational spaces to maintain a formal record of the non-disclosable owner(s) and/or producer(s) within an XML resource, if the resource is rendered for display within the protected internal organizational spaces in any format by a stylesheet or as a result of any other transformation process, then the non-disclosable owner(s) and/or producer(s) should not be included in the corresponding portion marking or banner marking. SHARED SPACES — Within shared spaces, the data element serves only to indicate the presence of FGI; in this case, this element's value will always be "FGI". The data element may also be employed in this manner within protected internal organizational spaces. Permissible values for this attribute are defined in the ISM FGI Protected CVE: CVEnumISMFGIProtected.xml. Note: the value "FGI" is permitted in the case outlined for "shared spaces".
{edh:ExternalEDH, edh:ResponsibleEntity, ism:SecurityAttriburesOptionGroup} ism:joint	When true, an indicator that entities in the @ism:ownerProducer attribute are JOINT owners of the data.
{edh:ExternalEDH, edh:ResponsibleEntity, ism:SecurityAttributesOptionGroup} ism:Nonicmarkings	One or more indicators of an expansion or limitation on the distribution of a document or portion originating from non-intelligence components This attribute is used at both the resource and the portion levels, and it is rendered in portion marks and security banners. The permissible values for this attribute are defined in the ISM Non-IC CVE: CVEnumISMNonIC.xml
{edh:ExternalEDH, edh:ResponsibleEntity, ism:SecurityAttributesOptionGroup} ism:Nonuscontrols	One or more indicators of an expansion or limitation on the distribution of a document or portion originating from non-US components (foreign government or international organization). This attribute is used at both the resource and the portion levels, and it is rendered in portion marks and security banners. The permissible values for this attribute are defined in the ISM Non-US Controls CVE: CVEnumISMNonUSControls.xml

IC-EDH	
EDH Element	Description
{edh:ExternalEDH,edh:ResponsibleEntity,ism:SecurityAttributesOptionGroup} ism:OwnerProducer	The set of national governments and/or international organizations that have purview over the containing classification marking This element is always used in conjunction with the Classification element. Taken together, the two elements specify the classification category (TS, S, C, R, or U) and the type of classification (US, non-US, or Joint). The permissible values for this attribute are defined in the ISMCAT Owner Producer CVE: CVEnumISMCATOwnerProducer.xml.
{edh:ExternalEDH,edh:ResponsibleEntity,ism:SecurityAttributesOptionGroup} ism:releasableTo	This attribute is used at both the resource and the portion levels. One or more indicators identifying the country or countries and/or international organization(s) to which classified information may be released based on the determination of an originator in accordance with established foreign disclosure procedures. This element is used in conjunction with the Dissemination Controls element. It is manifested in portion marks and security banners. PERMISSIBLE VALUES The permissible values for this attribute are defined in the Controlled Value Enumeration: CVEnumISMRelTo.xml
{edh:ExternalEDH,edh:ResponsibleEntity,ism:SecurityAttributesOptionGroup} ism:SARIdentifier	This attribute is used at both the resource and the portion levels. One or more indicators identifying the defense or intelligence programs for which special access is required. It is manifested in portion marks and security banners. PERMISSIBLE VALUES The permissible values for this attribute are defined in the Controlled Value Enumeration: CVEnumISMSAR.xml
{edh:ExternalEDH, edh:ResponsibleEntity} ism:SecurityAttributesOptionGroup	This attribute is used at both the resource and the portion levels. One or more indicators identifying sensitive compartmented information control system(s). It is manifested in portion marks and security banners. PERMISSIBLE VALUES The permissible values for this attribute are defined in the Controlled Value Enumeration: CVEnumISMSCIControls.xml
{edh:ExternalEDH,edh:ResponsibleEntity} edh:Country	The ISO 3166 country code of the organization that created the data item.
{edh:ExternalEDH,edh:ResponsibleEntity} edh:Organization	The organization within the country specified that created the data item.
{edh:ExternalEDH,edh:ResponsibleEntity} edh:Suborganization	The sub organization within the organization specified that created the data item.
{edh:ExternalEDH} edh:DataSet	A means of indicating a particular association of data. MUST not impact access control, access control MUST be entirely contained in the ARH section.
{edh:ExternalEDH,arh:ExternalSecurity} ism:compliesWith	e.g. compliesWith="USGov USIC
{edh:ExternalEDH,arh:ExternalSecurity} ism:resourceElement	e.g. TRUE
{edh:ExternalEDH,arh:ExternalSecurity} ism:createDate	e.g. 5/4/206
{edh:ExternalEDH,arh:ExternalSecurity} ism:classification	e.g. U

IC-EDH	
EDH Element	Description
{edh:ExternalEDH,arh:ExternalSecurity} ism:ownerProducer	e.g. USA
{edh:ExternalEDH,arh:ExternalSecurity} ism:excludeFromRollup	e.g. TRUE
{edh:ExternalEDH,ntk:ExternalAccess}	A way of describing a formalized Need to Know required for a document. NTK requires being inside a schema that implements ISM therefore some element in the implementing schema MUST have ism:ISMRootNodeAttributeGroup and ism:ResourceNodeAttributeGroup since both of those are required for a valid ISM implementation. In addition the root node of the implementing schema must have ntk:NTKRootNodeAttributeGroup specified
{edh:ExternalEDH,ntk:ExternalAccess} ntk:externalReference	e.g. TRUE
{edh:ExternalEDH,ntk:ExternalAccess} ism:classification	e.g. U
{edh:ExternalEDH,ntk:ExternalAccess} ism:ownerProducer	e.g. USA
{edh:ExternalEDH,ntk:ExternalAccess} ntk:AccessProfile	An access policy name followed by 1 or more values representing a profile that a user wanting access to the document must meet. Logic for the profile list being Boolean AND vs OR is Access Policy defined.
{edh:ExternalEDH,ntk:ExternalAccess} ntk:RequiresAllOf	A logical grouping indicating that all included access policies are required.
{edh:ExternalEDH,ntk:ExternalAccess} ntk:RequiresAnyOf	A logical grouping indicating that one of the included access policies is required.
{edh:ExternalEDH,ntk:ExternalAccess, ism:NoticeExternal}	A single Notice that may consist of 1 or more NoticeText for use when the notice refers to something external.
{edh:ExternalEDH,ntk:ExternalAccess, ism:NoticeExternal} ism:externalNotice	The actual notice of a text.

APPENDIX F: NGA METADATA PROFILE

The following Table is a sequential output of all the data elements are represented as NGA required fields and IC required standards in order or relevance:

- National Metadata Foundation (NMF)
- Additional per feedback per CDO, A, S, and X
- Electronic Rights Management (ERM)
- Trusted Data Format (TDF)
- IC Enterprise Data Header (IC EDH)

NGA Metadata Profile		
Metadata Point of Contact	NMF 3.0	The party or parties acting in a role of responsibility for a set of resource metadata.
Metadata Date	NMF 3.0	The date(s), and optionally time(s), of an event involving a set of resource metadata.
Metadata Standard Title	NMF 3.0 A&S	The name of the metadata standard which determines the structure and content of this set of resource metadata.
Metadata Standard Edition	NMF 3.0 A&S	The version of the metadata standard which determines the structure and content of this set of resource metadata.
Metadata Scope Code	NMF 3.0	A brief indicator of the type of resource for which metadata information is reported.
Metadata Scope Name	NMF 3.0	A word or phrase that describes the type of resource for which information is reported.
Metadata Classification	NMF 3.0	The classification level of the metadata, in accordance with the Intelligence Community (IC) Security Markings Manual.
Metadata Classification System	NMF 3.0	The classification system of the metadata, in accordance with the Intelligence Community (IC) Security Markings Manual.
Resource Title	NMF 3.0	The name by which a cited resource is known.
Resource Abstract	NMF 3.0 A&S X	A brief statement or narrative summary of the resource.
{edh:ExternalEDH}edh:ResponsibleEntity {NMF} Resource Point of Contact	EDH NMF 3.0 A&S X	The party(ies) acting in a role of responsibility for the resource.
{edh:ExternalEDH}edh:DateItemCreateDate {NMF} Resource Date	EDH NMF 3.0 A&S	A reference date for a cited resource. The first instance of this element is to have the role of "creation".
{edh:ExternalEDH}icid:Identifier {NMF} Resource Identifier	NMF 3.0 EDH A&S	A value uniquely identifying the resource within a namespace.

NGA Metadata Profile		
Resource Geographic Location	NMF 3.0 A&S	The spatial extent of the resource. The spatial extent is a geographic identifier (for example: a country name), a bounding box (for example: the bounding latitudes and longitudes), or a bounding object (for example: a set of coordinate points).
Resource Language	NMF 3.0 A&S	Designation of the locale language.
Resource Character Set	NMF 3.0	Designation of the character set to be used to encode the textual value of the locale.
Resource Topic Category Code	NMF 3.0 A&S X	A theme or topic keyword that represents a subject of the resource.
Resource Keywords	NMF 3.0 A&S X	Information about keywords describing this resource.
{edh:ExternalEDH, edh:ResponsibleEntity, ism:SecurityAttributesOptionGroup}ism:classification {NMF} Resource Classification	EDH NMF 3.0	The classification level of the resource, in accordance with the Intelligence Community (IC) Security Markings Manual.
{edh:ExternalEDH, edh:ResponsibleEntity, ism:SecurityAttributesOptionGroup}ism:derivedFrom {NMF} Resource Classification System	EDH NMF 3.0 A&S	The classification system of the resource, in accordance with the Intelligence Community (IC) Security Markings Manual.
Resource Category	NMF 3.0	The particular category of a data resource within a defined taxonomy.
Metadata Language	NMF 3.0	Designation of the locale language.
Metadata Character Set	NMF 3.0	Designation of the character set to be used to encode the textual value of the locale
Parent Metadata Citation	NMF 3.0	A standardized reference to a set of metadata that is in a parent relationship to this set of resource metadata
Resource Temporal Extent	NMF 3.0 A&S	The time period covered by the resource
Resource Coordinate Reference System	NMF 3.0	Information about a spatial or temporal reference system used by representations in the resource.
Content-ID	NGA CDO	Number designated by the NGA to label content type.
Maintenance Cycle/Update Frequency	X	Indicates how often the object is updated
Online Resource	A&S X	information about on-line sources from which the dataset, can be obtained (hyperlink to object)
erm:AppliedBy	ERM	Individual that performed the disposition of the Record.
erm:Authorizer	ERM	Entity responsible for the hold.
erm:DateApplied	ERM	Date when action was taken or applied to the Record.
erm:DateEligible	ERM	Date the record is eligible for disposition.

<i>NGA Metadata Profile</i>		
erm:DateLimit	ERM	Date by which a disposition action must be taken regarding the Record.
erm:Dispositoin	ERM	An action taken with regard to U.S. federal records that are no longer needed for current government business as determined by their appraisal pursuant to legislation, regulation, or administrative procedure.
erm:EffectiveDate	ERM	Date the hold order was effective.
erm:ElectronicRecords-ManagementMetadata	ERM	Information used by a records management system to manage a resource. A records management system systematically controls the creation, distribution, use, maintenance, and disposition of recorded information maintained as evidence of business activities and transactions.
erm:FoiaOpsIndicator	ERM	Indicator stating if the Record is exempted from FOIA search and publication (e.g. operations exemption at CIA): each agency has specific exemptions they request and maintain.
erm:Hold	ERM	A suspension of disposition of a set of one or more records in the records management process of the cognizant organization or institution.
erm:Identifier	ERM	Specific way to identify the hold (unique among hold types and agencies).
erm:Justification	ERM	Legal, policy, or mission driver for the hold order.
erm:OfficeOfRecord	ERM	Identifies the Agency and organization element within the agency that is responsible for making decisions related to the Record. Also identifies which organizational element within an agency owns/is responsible for the official copy of the Record.
erm:RecordControl	ERM	Identifier for Agency Record Control System (RCS) disposition policies (per NARA).
erm:RecordDesignationDate	ERM	Date the Record is declared final (e.g. cutoff, publication, creation) and starts the retention period which will be used to calculate the disposition date based on the retention schedule.
erm:ReleasedDate	ERM	Date the hold order was released.
erm:ReviewIndicator	ERM	Indicator stating that the resource has been reviewed to determine the appropriate RCS, if applicable.
erm:Type	ERM	The category of a hold on a managed Record's disposition.
erm:VitalRecordsIndicator	ERM	Indicator stating that the Record is a Vital Record. Vital Records are those Records considered essential to the continuity of operation during and after emergencies or disaster conditions. Also known as Essential Records (per Federal Continuity Directive 1 (FCD 1))
{tdf} IC-TDF-collection	TDF	The root element of a Trusted Data Collection. A TDC can have a single HandlingAssertion containing and IC-EDH specification and its scope must be TDC. There may also be an optional second HandlingAssertion scope [TDC] that contains Revision/Recall information for the TDC using the RevRecall.XML specification. A HandlingAssertion must not be encrypted.
{TrustedDataCollection} tdf:HandlingAssertion	TDF	A specific type of assertion designed to be used for access, rights, and handling instructions. It is expected that handling instructions should never have metadata about themselves and they should never be encrypted. Therefore, unlike regular assertions, handling assertions do not support statement metadata or encryption.

<i>NGA Metadata Profile</i>		
{TrustedDataCollection, tdf:HandlingAssertion} attr:scope	TDF	The grouping of objects to which the assertion applies.
{TrustedDataCollection, tdf:HandlingAssertion} tdf:HandlingStatement	TDF	Intended for access, rights, and/or handling instructions that apply to the scope of the assertion.
{tdf} TrustedDataObject	TDF	A TDO has at a minimum two HandlingAssertions: A TDO handling assertion and a payload handling assertion. This allows for separate access control decisions to be made for the payload versus the entire TDO (which includes payload metadata).
{TrustedDataObject} tdf:HandlingAssertion	TDF	A specific type of assertion designed to be used for access, rights, and handling instructions. It is expected that handling instructions should never have metadata about themselves and they should never be encrypted. Therefore, unlike regular assertions, handling assertions do not support statement metadata or encryption.
{TrustedDataObject} tdf:Assertion	TDF	Used to express metadata about the objects expressed in the scope attribute of the assertion. An assertion also supports metadata about the assertion statement for the purposes of indicating any handling instructions pertain to the statement itself. Also supports encrypted statements and binding the statement with objects in its scope.
{TrustedDataObject, tdf:Assertion} attr:scope	TDF	The grouping of objects to which the assertion applies.
{TrustedDataObject, tdf:EncryptionInformationGroup} tdf:EncryptionInformation	TDF	(U) Top level element for holding information related to the encryption of an assertion or payload. Multiple child KeyAccess and/or EncryptionMethod elements represent onion or layered encryption. In this case, the first child represents the outermost layer of encryption. WILL USE ALL INFORMATION IF NECESSARY.
{TrustedDataObject, tdf:Assertion} tdf:ReferenceValuePayload	TDF	Used to reference payloads that are not embedded in the TDO but stored in a remote/external location.
{edh:ExternalEDH} edh:ExternalEDH	EDH	Used for specifying an object's basic header information required for exchange on the IC Enterprise when the object is not in the instance document that this element is used. Values specified in this header will not contribute to roll-up.
{edh:ExternalEDH} ntk:DESVersion	EDH	e.g. 201508
{edh:ExternalEDH} ism:DESVersion	EDH	e.g. 201508
{edh:ExternalEDH} ism:ISMATESVersion	EDH	e.g. 20105
{edh:ExternalEDH} arh:DESVersion	EDH	e.g. 3
{edh:ExternalEDH} edh:DESVersion	EDH	e.g. 201508
{edh:ExternalEDH} icid:Identifier or {NMF} Resource Identifier	EDH NMF 3.0 A&S	e.g. guide://
{edh:ExternalEDH} edh:DatatemCreateDate or {NMF} Resource Date	EDH NMF A&S	e.g. 2006-05-04T18:13:51.0Z

NGA Metadata Profile		
{edh:ExternalEDH} edh:AuthorizationReference	EDH	Attribute to hold the authorization reference of the data object referred to by the EDH. One or more indicators of a unique and documented legal basis for all activities surrounding the creation, retention and use of an information resource.
{edh:ExternalEDH} edh:AuthorizationReferenceType	EDH	One or more indicators of a unique and documented legal basis for all activities surrounding the creation, retention and use of an information resource.
{edh:ExternalEDH} edh:ResponsibleEntity or {NMF} Resource Point of Contact	EDH NMF A&S X	The creator entity that created the data item referenced by the EDH.
{edh:ExternalEDH, edh:ResponsibleEntity,ism:SecurityAttriburesGroup} ism:atomicEnergyMarkings	EDH	Applicable atomic energy information markings for a document or portion
{edh:ExternalEDH, edh:ResponsibleEntity,ism:SecurityAttriburesGroup} ism:classification or {NMF} Resource Classification	EDH NMF	The highest level of classification applicable to the containing document or portion
{edh:ExternalEDH, edh:ResponsibleEntity,ism:SecurityAttriburesGroup} ism:classificationReason	EDH	One or more reason indicators or explanatory text describing the basis for an original classification decision
{edh:ExternalEDH, edh:ResponsibleEntity,ism:SecurityAttributesOptionGroup} ism:classifiedBy	EDH	The identity, by name or personal identifier and position title, of the original classification authority for a document
{edh:ExternalEDH, edh:ResponsibleEntity,ism:SecurityAttributesOptionGroup} ism:compilationReason	EDH	The reason that the classification of the document is more restrictive than the simple roll-up of the marked portions of the document
{edh:ExternalEDH, edh:ResponsibleEntity,ism:SecurityAttributesOptionGroup} ism:declassDate	EDH	The specific date when the resource is subject to automatic declassification procedures if not properly exempted from automatic declassification
{edh:ExternalEDH, edh:ResponsibleEntity,ism:SecurityAttributesOptionGroup} ism:declassEvent	EDH	A description of an event upon which the information shall be subject to automatic declassification procedures if not properly exempted from automatic declassification A description of an event upon which the information shall be automatically declassified if not properly exempted from automatic declassification.
{edh:ExternalEDH, edh:ResponsibleEntity,ism:SecurityAttributesOptionGroup} ism:declassException	EDH	The exemption from automatic declassification that is claimed for a document
{edh:ExternalEDH, edh:ResponsibleEntity,ism:SecurityAttributesGroup}	EDH	The identity, by name or personal identifier, of the derivative classification authority

NGA Metadata Profile		
ism:derivativelyClassifiedBy		
{edh:ExternalEDH, edh:ResponsibleEntity,ism:SecurityAttriburesGroup} ism:derivedFrom {NMF} Resource Classification System	EDH NMF	A citation of the authoritative source or sources of the classification markings used in a derivative classification decision for a classified document
{edh:ExternalEDH, edh:ResponsibleEntity,ism:SecurityAttriburesGroup} ism:displayOnlyTo	EDH	The set of countries and/or international organizations associated with a “Display Only To” marking
{edh:ExternalEDH,edh:ResponsibleEntity,ism:SecurityAttributesOptionGroup} ism:disseminationControls	EDH A&S	Applicable dissemination control markings for a document or portion
{edh:ExternalEDH,edh:ResponsibleEntity,ism:SecurityAttributesOptionGroup} ism:FGISourceOpen	EDH	The set of countries and/or international organizations whose information is derivatively sourced in a document when the source of the information is not concealed (also used for cases when the source is unknown)
{edh:ExternalEDH, edh:ResponsibleEntity, ism:SecurityAttriburesOptionGroup} ism:FGISourceProtected	EDH	The set of countries and/or international organizations whose information is derivatively sourced in a document when the source of the information must be concealed
{edh:ExternalEDH, edh:ResponsibleEntity, ism:SecurityAttriburesOptionGroup} ism:joint	EDH	When true, an indicator that entities in the @ism:ownerProducer attribute are JOINT owners of the data
{edh:ExternalEDH,edh:ResponsibleEntity,ism:SecurityAttributesOptionGroup} ism:Nonicmarkings	EDH	One or more indicators of an expansion or limitation on the distribution of a document or portion originating from non-intelligence components This attribute is used at both the resource and the portion levels, and it is rendered in portion marks and security banners. The permissible values for this attribute are defined in the ISM Non-IC CVE: CVEEnumISMNonIC.xml
{edh:ExternalEDH,edh:ResponsibleEntity,ism:SecurityAttributesOptionGroup} ism:Nonuscontrols	EDH	One or more indicators of an expansion or limitation on the distribution of a document or portion originating from non-US components (foreign government or international organization).
{edh:ExternalEDH,edh:ResponsibleEntity,ism:SecurityAttributesOptionGroup} ism:OwnerProducer	EDH	The set of national governments and/or international organizations that have purview over the containing classification marking

NGA Metadata Profile		
{edh:ExternalEDH,edh:ResponsibleEntity,ism:SecurityAttributesOptionGroup} ism:releasableTo	EDH A&S	This attribute is used at both the resource and the portion levels. One or more indicators identifying the country or countries and/or international organization(s) to which classified information may be released based on the determination of an originator in accordance with established foreign disclosure procedures. This element is used in conjunction with the Dissemination Controls element. It is manifested in portion marks and security banners. PERMISSIBLE VALUES The permissible values for this attribute are defined in the Controlled Value Enumeration: CValueEnumISMRelTo.xml
{edh:ExternalEDH,edh:ResponsibleEntity,ism:SecurityAttributesOptionGroup} ism:SARIdentifier	EDH	This attribute is used at both the resource and the portion levels. One or more indicators identifying the defense or intelligence programs for which special access is required. It is manifested in portion marks and security banners. PERMISSIBLE VALUES The permissible values for this attribute are defined in the Controlled Value Enumeration: CValueEnumSMSAR.xml
{edh:ExternalEDH, edh:ResponsibleEntity} ism:SecurityAttributesOptionGroup	EDH	This attribute is used at both the resource and the portion levels. One or more indicators identifying sensitive compartmented information control system(s). It is manifested in portion marks and security banners. PERMISSIBLE VALUES The permissible values for this attribute are defined in the Controlled Value Enumeration: CValueEnumMSCIControls.xml
{edh:ExternalEDH,edh:ResponsibleEntity} edh:Country	EDH	The ISO 3166 country code of the organization that created the data item.
{edh:ExternalEDH,edh:ResponsibleEntity} edh:Organization	EDH	The organization within the country specified that created the data item.
{edh:ExternalEDH,edh:ResponsibleEntity} edh:Suborganization	EDH	The suborganization within the organization specified that created the data item.
{edh:ExternalEDH} edh:DataSet	EDH	A means of indicating a particular association of data. MUST not impact access control, access control MUST be entirely contained in the ARH section.
{edh:ExternalEDH,arh:ExternalSecurity} ism:compliesWith	EDH	e.g. compliesWith="USGov USIC
{edh:ExternalEDH,arh:ExternalSecurity} ism:resourceElement	EDH	e.g. TRUE
{edh:ExternalEDH,arh:ExternalSecurity} ism:createDate	EDH	e.g. 5/4/206
{edh:ExternalEDH,arh:ExternalSecurity} ism:classification	EDH	e.g. U
{edh:ExternalEDH,arh:ExternalSecurity} ism:ownerProducer	EDH	e.g. USA
{edh:ExternalEDH,arh:ExternalSecurity} ism:excludeFromRollup	EDH	e.g. TRUE

NGA Metadata Profile		
{edh:ExternalEDH,ntk:ExternalAccess}	EDH	A way of describing a formalized Need to Know required for a document. NTK requires being inside a schema that implements ISM therefore some element in the implementing schema MUST have <code>ism:ISMRootNodeAttributeGroup</code> and <code>ism:ResourceNodeAttributeGroup</code> since both of those are required for a valid ISM implementation. In addition the root node of the implementing schema must have <code>ntk:NTKRootNodeAttributeGroup</code> specified
{edh:ExternalEDH,ntk:ExternalAccess} ntk:externalReference	EDH	e.g. TRUE
{edh:ExternalEDH,ntk:ExternalAccess} ism:classification	EDH	e.g. U
{edh:ExternalEDH,ntk:ExternalAccess} ism:ownerProducer	EDH	e.g. USA
{edh:ExternalEDH,ntk:ExternalAccess} ntk:Access-Profile	EDH	An access policy name followed by 1 or more values representing a profile that a user wanting access to the document must meet. Logic for the profile list being Boolean AND vs OR is Access Policy defined.
{edh:ExternalEDH,ntk:ExternalAccess} ntk:RequiresAllOf	EDH	A logical grouping indicating that all included access policies are required.
{edh:ExternalEDH,ntk:ExternalAccess} ntk:RequiresAnyOf	EDH	A logical grouping indicating that one of the included access policies is required.
{edh:ExternalEDH,ntk:ExternalAccess, ism:NoticeExternal}	EDH	A single Notice that may consist of 1 or more NoticeText for use when the notice refers to something external.
{edh:ExternalEDH,ntk:ExternalAccess, ism:NoticeExternal} ism:externalNotice	EDH	The actual notice of a text.

APPENDIX G: DATA IN THE CLOUD – EXAMPLE CASES

This Appendix presents two example paths for activities evolving into the cloud. These example use cases are to assist Data Stewards and Program Managers design a customized appropriate path for their own data. Both examples move from a system architecture where data is tightly coupled to the applications (activities) to an architecture where the data is decoupled from the applications through the introduction of data services.

Example One - describes a potential path to evolve from a system that manages data objects with integrated (i.e., tightly coupled to the application) metadata management and access controls to a set of multi-purpose enterprise data management and access and security services, using S3 as the storage service.

Example Two - takes a database that uses Stateful client-server technology to a cloud (C2S RDS) database using RESTful services. Both paths balance moving the data to the cloud with minimizing the near term impact to the applications.

Representative activities are used to allow for more identifiable examples. But it should be understood that the details are “notional” and NCL and AMS continue to define their evolution paths as part of their respective cloud migration efforts.

G.1 Data Storage Architecture Evolution – Object Storage (S3) Example Case

An object storage example, using the NSG Consolidated Library (NCL) illustrates data architecture evolution where the main data are objects – specifically, a large library of imagery. NCL has its own catalog of imagery metadata, its own sophisticated policy enforcement capabilities which can enforce access down to the pixel block level, its own image format, and transformation services.

NOTE: DIAGRAMS ARE NOTIONAL - NOT MEANT TO REPRESENT ACTUAL PHYSICAL CONFIGURATIONS

Now – Current State

In the current state, the OpenDataStore exists, but there are no PEPs defined, no established enterprise Metadata Catalog, and no enterprise Image Format Transformation services.

As shown in Figure H-1, the NCL imagery is migrated to S3 “Dirty Buckets” in the OpenDataStore. Access to NCL imagery would still require access through the NCL application.

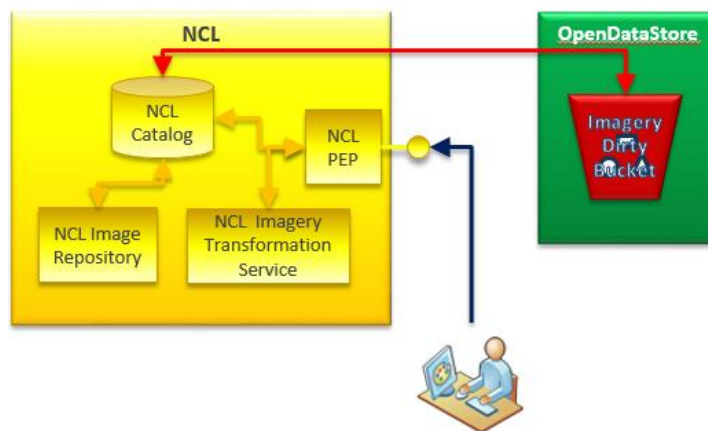


Figure G-1: S3 Now – Current State Architecture

Next – Transitional State

In the transition state, NCL imagery in the “Dirty Buckets” will be conditioned (i.e., metadata tagged) and registered in the enterprise metadata catalog, which changes its status to a useable “clean” state. At this point, NCL and non-NCL users will be able to discover the imagery which has been conditioned and registered in the enterprise metadata catalog without using the NCL application. See metadata guidance for more details on conditioning and the enterprise metadata catalog.

As shown in Figure H-2, NCL would operate as it had previously with the exception of having the data stored in the OpenDataStore.

Users could still use the NCL activity and its suite of services if desired, but with the following services implemented would not need to, taking the first step to truly separating the application from the data. Enterprise services would include:

- PEP will be defined. Business rules will be established to negotiate user access rights and data/metadata attributes. Initially enforcement may only go to the file level, instead of pixel block level enforcement, but will allow access control without using the NCL application.
- Enterprise Imagery Format Transformation services. This could take any imagery in any format it supports and transform it to any other imagery format supported.

Target State

In the target state, the enterprise services would achieve pixel block level policy enforcement, as available in NCL today.

As shown in Figure H-3, NCL would transition from using its own PEP to the Metadata Catalog and Imagery Format Transformation suite of enterprise services instead. What remains in NCL would be services unique to it, such as the user interface and reports.

One architectural objective is to separate the data from the activity. To this end, the architecture includes the enterprise services needed to discover, access and transform the data for use by anyone or activity in the NSG.

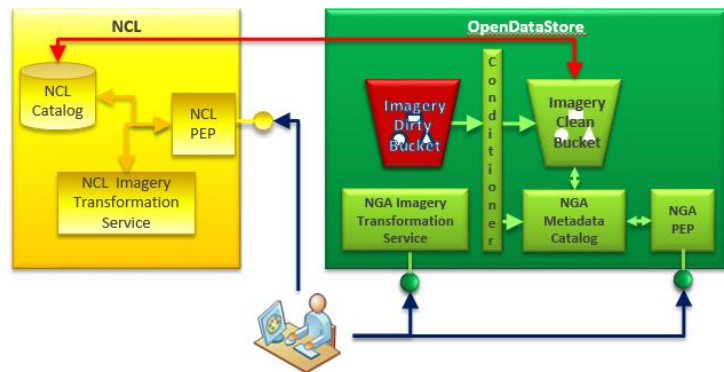


Figure G-2: S3 Next - Transitional State

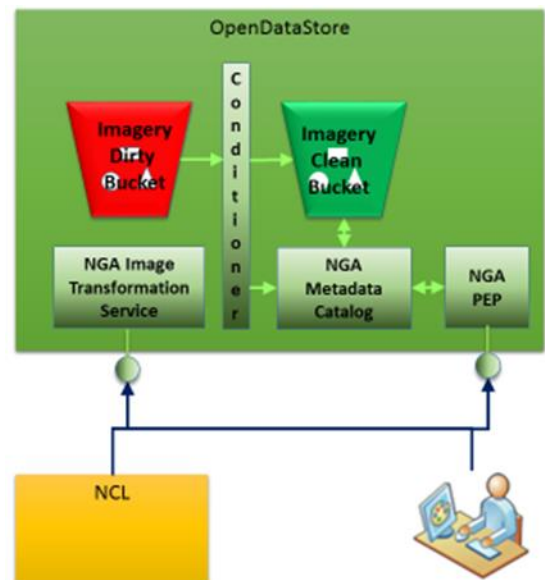


Figure G-3: Cloud OpenDataStore, S3 Target State

G.2 Data Storage Architecture Evolution – Database (RDB) Example Case

The Aeronautical Migration System (AMS) is the production system for Aeronautical Safety of Navigation (SoN). It is subject to intense international regulations to enable the data to be used in aircraft Flight Management Systems (FMS, a.k.a. Autopilot). To maintain their current certification, they are audited every 6 months. If they change technologies significantly, it will require a completely new certification, which could take over a year after the new development is completed.

The current AMS system, shown in Figure H-4, is client-server based, with Sybase databases, Visual Basic 5 & 6 clients, and a Boundless OpenGeo QGIS client. The Visual Basic clients connect via ODBC directly to the database. The OpenGeo QGIS client connects via an OGC Transactional Web Feature Service (WFS-T). Visual Basic clients need to be maintained to support non-spatial data.

Option 1- Re-host Sybase

The easiest cloud migration option is re-hosting - setting up an EC2/EBS instance in the OpenDataStore and recreate the existing Sybase databases, as shown in Figure H-5. This maintains legacy functionality without triggering a recertification of AMS because the software technology stack is unchanged. However, other than getting AMS out of the NGA Data Center, substantial benefits are unlikely.

Option 2: Migrate to RDS

Aeronautical has been contemplating updating to a more modern database and identified PostgreSQL with PostGIS as the best candidate. They intend to consolidate their Sybase databases into a unified C2S RDS PostgreSQL database. This transformation, as shown in Figure H-6, allows transitioning away from the ODBC connection, and represents a significant cost savings in licensing costs (i.e., PostgreSQL is open source) and infrastructure maintenance resources (i.e., use of C2S RDS). However, they cannot use the new system to produce Aeronautical SoN products without recertifying the system, which will take over a year after development is complete.

Option 3: Phased Combined Solution

Aeronautical can do a two pronged approach. They lift and shift their current system to meet current cloud mandate.

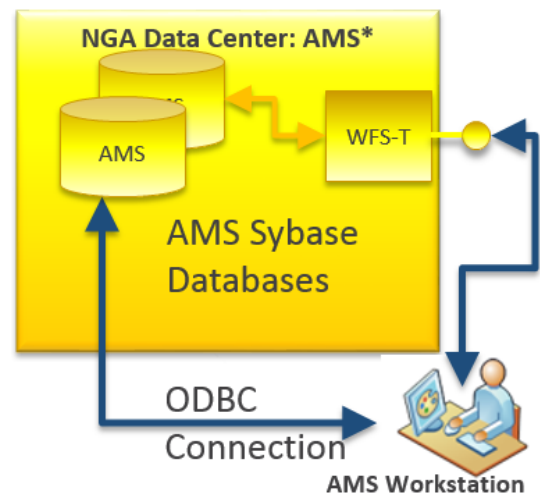


Figure G-4: Database As-Is Example

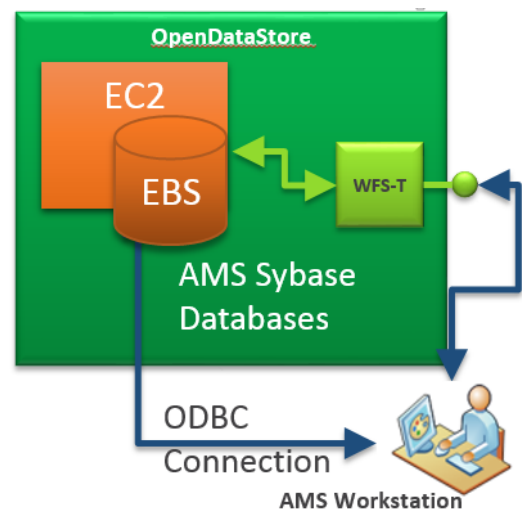


Figure G-5: Option 1 - Lift & Shift Rehost

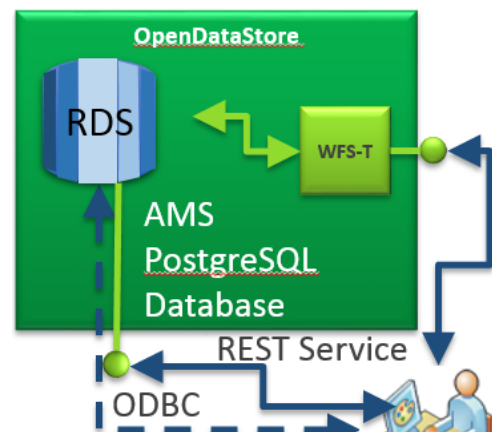


Figure G-6: Option 2 RDS Solution

At the same time, they develop their new capabilities in the cloud, using cloud native technologies. As soon as the new system is certified, they release the EC2/EBS instances used for the old system.

Eventual Target

The initial migration of NGA databases preserves the original schema and interfaces of the legacy activities. Over time, as shown in Figure H-7, the direct connect interfaces like the ODBC interfaces will be replaced with web based services and APIs, supporting loosely-coupled solutions and enable Object Based Production (OBP).

Once this is accomplished, and the data is abstracted from the activities, database consolidation can occur behind the scenes, as long as the published interfaces for the activities are maintained. Databases like the Aeronautical SoN database could be consolidated into an overall OBP database that would contain all NGS feature/object data.

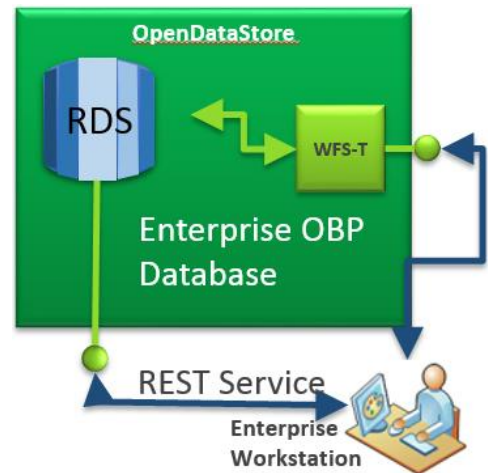


Figure G-7: Eventual Target Cloud Database

APPENDIX H: ODS S3 BUCKET NAMING STANDARD

S3 Buckets Naming Standard			
<i>Data Category</i>	<i>Government</i>	<i>Commercial</i>	<i>Host Nation</i>
Geodesy	Geodesy-Government	Geodesy-Commercial	Geodesy-HostNation
Human Geography	HumanGeography-Government	HumanGeography-Commercial	HumanGeography-HostNation
Aeronautical	Aeronautical-Government	Aeronautical-Commercial	Aeronautical-HostNation
Maritime	Maritime-Government	Maritime-Commercial	Maritime-HostNation
Topographic Features	Topographic-Government	Topographic-Commercial	Topographic-HostNation
Targeting	Targeting-Government	Targeting-Commercial	Targeting-HostNation
Names & Boundaries	NamesBoundaries-Government	NamesBoundaries-Commercial	NamesBoundaries-HostNation
Controlled Imagery	ControlledImagery-Government	ControlledImagery-Commercial	ControlledImagery-HostNation
Elevation	Elevation-Government	Elevation-Commercial	Elevation-HostNation
Imagery	Imagery-Government	Imagery-Commercial	Imagery-HostNation
Intelligence Report	IntelligenceReport-Government	IntelligenceReport-Commercial	IntelligenceReport-HostNation
Knowledge	Knowledge-Government	Knowledge-Commercial	Knowledge-HostNation
Corporate	Corporate-Government	Corporate-Commercial	Corporate-HostNation
Unknown	Unknown-Government	Unknown-Commercial	Unknown-HostNation

S3 bucket with datasets mapped to it in the AR
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